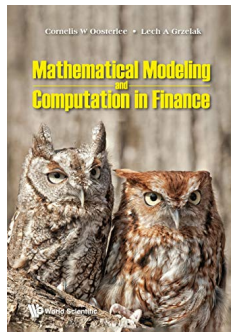


Materials for the course

The course is based on book

“*Mathematical Modeling and Computation in Finance:
With Exercises and Python and MATLAB Computer Codes*”,
by C.W. Oosterlee and L.A. Grzelak, WSP Europe Ltd, 2019.



- ▶ For more details regarding the book go [here](#).
- ▶ YouTube Channel with courses can be found [here](#).
- ▶ Slides and the codes can be found [here](#).

Why This Series Exists

Quantitative finance uses sophisticated mathematics, statistics, optimization and computing. Yet some of the most costly failures do not arise from difficult mathematics. They arise from **violating** simple principles.

Typical examples include:

- ▶ using information unavailable at the decision time;
- ▶ confusing correlation with causation;
- ▶ overfitting historical data;
- ▶ ignoring model assumptions;
- ▶ mistaking simulation output for economic reality;
- ▶ confusing precision with accuracy.

Objective of the series

- ▶ The purpose is not merely to catalogue mistakes. The objective is to build a systematic framework for recognizing, diagnosing and preventing them.

Central idea

- ▶ *Many quant disasters originate from a simple conceptual error hidden inside a sophisticated model.*

What Is a Quant Mistake?

Each episode starts from a statement that appears reasonable, but is either mathematically incorrect or financially misleading.

Incorrect Belief $\longrightarrow$ Quant Mistake $\longrightarrow$ Incorrect Decision

The incorrect belief

- ▶ A plausible idea that many practitioners might accept without careful examination.

The mistake

- ▶ *A violation of a mathematical, statistical or financial principle.*

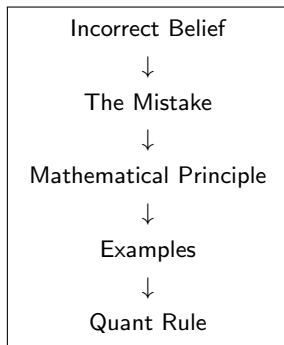
The consequence

- ▶ Incorrect prices, unstable hedges, misleading risk measures, unrealistic backtests or poor investment decisions.

The first goal is always to understand exactly why the mistake appears attractive before explaining why it is wrong.

How Each Episode Is Structured

Every episode follows the same sequence.



- ▶ The intuition is presented first.
- ▶ The mathematics identifies the violated principle.
- ▶ Practical examples show how the mistake appears in real work.
- ▶ A Quant Rule summarizes the lesson.

Why use a fixed structure?

- ▶ Consistency allows the audience to compare different mistakes and recognize recurring patterns across disciplines.

The Role of Examples

Understanding improves dramatically when abstract ideas are connected to practical situations. Each episode therefore uses two complementary types of examples.

Realistic finance example

- ▶ A pricing, hedging, calibration, risk-management or trading problem illustrates how the mistake appears in practice.

Counterexample

- ▶ *A deliberately extreme example exposes the flaw by producing an impossible result such as:*
 - ▶ *a trading strategy that never loses;*
 - ▶ *a perfect forecast;*
 - ▶ *a flawless calibration;*
 - ▶ *a risk model with zero uncertainty.*

Such examples often make the underlying error immediately visible.

What You Should Learn from This Series

The goal is not to memorize thirty isolated mistakes. The goal is to develop a repeatable way of thinking.

Recognize → Diagnose → Correct → Prevent

For every model, strategy or calculation, ask:

Which assumptions am I making?

Which principle justifies this step?

What could invalidate the result?

Final message

- ▶ *The most valuable lesson in quantitative finance is not a formula. It is the ability to recognize when a formula is being used incorrectly.*

The Biggest Mistakes in Quant Finance

Quant Mistakes 1–10

Risk-neutral pricing

1. Lecture FE2: Using information from the future.
2. Lecture FE2: Pricing derivatives under the wrong probability measure.
3. Lecture FE2: Changing the numeraire without changing the drift.
4. Lecture CF2: Applying ordinary calculus to stochastic processes.

Volatility misconceptions

5. Lecture CF4: Treating implied volatility as a forecast of realized volatility.
6. Lecture CF4: Trusting an implied volatility because the root-finder converged.
7. Lecture CF4: Using one Black–Scholes volatility for all strikes and maturities.
8. Lecture CF5: Assuming that delta hedging removes jump risk.
9. Lecture CF5–CF6: Adding jumps without compensating the drift.

The Biggest Mistakes in Quant Finance

Quant Mistakes 11–20

Model construction

- 10. Lecture CF6–CF7: Confusing calibration quality with model validity.
- 11. Lecture CF7: Assuming that stochastic volatility solves every smile problem.

Numerical implementation

- 12. Lecture CF8: Using Fourier pricing without checking numerical convergence.
- 13. Lecture CF9: Believing that more Monte Carlo paths make the result correct.
- 14. Lecture CF9: Confusing weak convergence with strong convergence.
- 15. Lecture CF10: Treating variance truncation as a harmless fix.
- 16. Lecture CF10: Assuming that a risk-neutral discretization preserves the martingale property.
- 17. Lecture CF11: Computing Monte Carlo Greeks with independent random numbers.
- 18. Lecture CF11: Assuming that delta hedging removes all financial risk.

Exotic derivatives

- 19. Lecture CF12: Assuming today's vanilla surface determines forward-start prices
- 20. Lecture CF13: Ignoring the path when pricing path-dependent derivatives.

The Biggest Mistakes in Quant Finance

Quant Mistakes 21–30

Interest-rate products

- 21. Lecture FE5, FE11: Treating a forward rate as a forecast of the future rate.
- 22. Lecture FE11: Treating a par swap rate as a zero rate.
- 23. Lecture FE6: Using one yield curve for both forecasting and discounting.
- 24. Lecture FE6: Choosing yield-curve interpolation only because it looks smooth.
- 25. Lecture FE7: Using a lognormal swap-rate model in a negative-rate environment.
- 26. Lecture FE11: Assuming that all forward rates are driftless under one measure.
- 27. Lecture FE11: Ignoring the convexity effect of a payment delay.
- 28. Lecture FE10: Pricing FX forwards without cross-currency basis spreads.

Risk management

- 29. Lecture FE12: Taking positive exposure before netting-set aggregation.
- 30. Lecture FE13: Treating Value-at-Risk as the maximum possible loss.

Quant Mistake #1: Using Information from the Future

The incorrect belief

- ▶ A pricing, hedging or trading decision may use information regardless of when that information becomes available.

The mistake

- ▶ *A decision made at time t uses information that becomes available only after time t . This mistake is also called look-ahead bias or data leakage.*

Typical examples:

- ▶ using the future stock value $S(T)$ at time $t < T$;
- ▶ using a floating-rate fixing before its fixing date;
- ▶ using tomorrow's return to determine today's trading position;
- ▶ using future observations when calibrating a historical model;
- ▶ using the full simulated path to make an earlier decision.

The consequence

- ▶ The resulting price, hedge or backtest may look exceptionally accurate, but it is not implementable and has no financial meaning.

Information Arrives over Time

Let $\mathcal{F}(t)$ denote all information available at time t .

- ▶ Information increases over time:

$$\mathcal{F}(t_0) \subseteq \mathcal{F}(t_1) \subseteq \cdots \subseteq \mathcal{F}(T).$$

- ▶ At time t , we may know

$$S(s), \quad 0 \leq s \leq t.$$

- ▶ We do not yet know

$$S(u), \quad u > t.$$

- ▶ Thus $S(t)$ is $\mathcal{F}(t)$ -measurable, whereas in general

$$S(T) \text{ is not } \mathcal{F}(t)\text{-measurable,} \quad T > t.$$

Important distinction

- ▶ *Knowing the probability distribution of a future quantity is not the same as knowing its realized future value.*

A simulation may generate an entire path numerically, but future values on that path are still unavailable to an earlier decision.

Learn more

- ▶ For filtrations, measurability and conditional expectations, see the [Financial Engineering course, Lecture FE2: Understanding of Filtrations and Measures.](#)

Pricing: Future Payoff versus Current Value

Let $H(T)$ be a derivative payoff at maturity T .

- ▶ The arbitrage-free value at time t is

$$V(t) = M(t) \mathbb{E}^{\mathbb{Q}} \left[\frac{H(T)}{M(T)} \mid \mathcal{F}(t) \right].$$

- ▶ Under a constant interest rate r ,

$$V(t) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}} [H(T) \mid \mathcal{F}(t)].$$

- ▶ The future payoff is therefore averaged over outcomes that are still possible given the information available at time t .
- ▶ It is generally incorrect to write

$$V(t) = e^{-r(T-t)} H(T), \quad t < T.$$

Why is this wrong?

- ▶ The uncertainty disappears only because the future realized payoff has been inserted into the calculation.
That is not pricing. It is discounting a payoff that becomes known later.

Example: A Future Floating-Rate Fixing

Consider, in a single yield curve setting, a floating coupon

$$H_i(T_i) = N\tau_i\ell(T_{i-1}; T_{i-1}, T_i),$$

where T_{i-1} is the fixing date and T_i is the pay date.

- ▶ Before fixing,

$$t < T_{i-1},$$

the realized fixing is unknown.

- ▶ The valuation therefore uses the forward rate available at time t :

$$V_i(t) = N\tau_i P(t, T_i)\ell(t; T_{i-1}, T_i).$$

- ▶ After fixing,

$$T_{i-1} \leq t < T_i,$$

the fixing has become known, and

$$V_i(t) = N\tau_i P(t, T_i)\ell(T_{i-1}; T_{i-1}, T_i).$$

The mistake

- ▶ Using the realized fixing before T_{i-1} inserts information from the future directly into today's valuation.

Learn more

- ▶ For floating-rate products, fixings and forward rates, see the [Financial Engineering course, Lecture FE5: Interest Rate Products](#).

Example: The Perfect Historical Backtest

Consider trading dates $t_0 < t_1 < \dots < t_m$. And the P&L is given by:

$$P\&L_i = \phi_i(S(t_{i+1}) - S(t_i))$$

Suppose the position, ϕ_i , at time t_i is chosen as

$$\phi_i = \text{sgn}(S(t_{i+1}) - S(t_i)).$$

Then

$$P\&L_i = |S(t_{i+1}) - S(t_i)| \geq 0.$$

- ▶ The strategy never loses.
- ▶ The backtest therefore appears perfect.
- ▶ But at time t_i , the value $S(t_{i+1})$ is not known.
- ▶ Tomorrow's return has been used to choose today's position.

A valid trading rule must depend only on information available at t_i :

$$\phi_i = g_i(S(t_0), \dots, S(t_i)).$$

Lesson

- ▶ *A spectacular backtest may reveal superior information about the future rather than a superior trading strategy.*

Quant Rule #1

Look-ahead bias often enters unintentionally through:

- ▶ revised economic data used in place of originally available data;
- ▶ normalization of the full data set before train–test splitting;
- ▶ calibration using observations after the valuation date;
- ▶ future fixings inserted before their fixing dates;
- ▶ full simulated paths used for earlier decisions.

For every input X , ask whether

$$t_{\text{obs}}(X) \leq t_{\text{decision}}.$$

Quant Rule #1

- ▶ At time t , every input to a pricing, hedging, exercise or trading decision must be available at time t .

Mathematically,

$$\sigma(\text{all inputs used at time } t) \subseteq \mathcal{F}(t).$$

Final message

- ▶ *A result that looks too good may not come from a better model. It may come from using information that was unavailable when the decision had to be made.*

Quant Mistake #2: Pricing Derivatives under the Wrong Probability Measure

The incorrect belief

- ▶ If the stock is expected to grow at rate μ , then a derivative can be priced by simulating

$$\frac{dS(t)}{S(t)} = \mu dt + \sigma dW^{\mathbb{P}}(t)$$

and discounting the average payoff at the risk-free rate.

The mistake

- ▶ *The real-world measure $\mathbb{P}$ describes expected market outcomes. It does not, by itself, determine arbitrage-free derivative prices:*

$$V(t) \neq M(t) \mathbb{E}^{\mathbb{P}} \left[\frac{H(T)}{M(T)} \mid \mathcal{F}(t) \right].$$

- ▶ $\mathbb{P}$ describes how we believe markets actually evolve.
- ▶ $\mathbb{Q}$ is consistent with no-arbitrage prices of traded assets.
- ▶ For a replicable claim, the price should not depend on a subjective estimate of the expected return μ .

Two Measures Answer Two Different Questions

Under the real-world measure,

$$\frac{dS(t)}{S(t)} = \mu dt + \sigma dW^{\mathbb{P}}(t).$$

Under the risk-neutral pricing measure,

$$\frac{dS(t)}{S(t)} = r dt + \sigma dW^{\mathbb{Q}}(t).$$

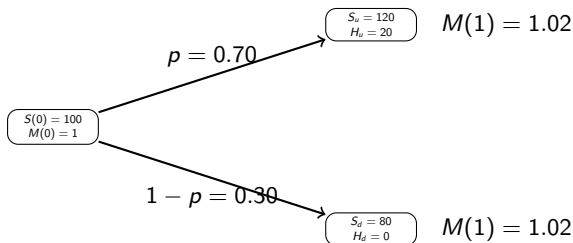
- ▶ Under $\mathbb{P}$, we ask:
 - ▶ What return do we expect?
 - ▶ What losses may occur?
 - ▶ How likely is a particular market scenario?
- ▶ Under $\mathbb{Q}$, we ask:
 - ▶ What price is consistent with traded assets?
 - ▶ What value avoids arbitrage?
 - ▶ What price is implied by replication?

Important

- ▶ *Risk-neutral pricing does not mean that investors truly expect stocks to grow at the risk-free rate.*

A Simple Example: The Wrong Price

- Consider a one-period market (Call option $H = \max(S(T) - K, 0)$ with $K = 100$):



- The naive physical-expectation price is

$$\begin{aligned}\tilde{V}^{\mathbb{P}}(0) &= \frac{1}{1.02} (0.70 \cdot 20 + 0.30 \cdot 0) \\ &= 13.725.\end{aligned}$$

The problem

- This “price” depends on the estimated physical probability p . But the option payoff can be replicated uniquely from traded assets. Its arbitrage-free price therefore cannot depend on our subjective estimate of p .*

The Risk-Neutral Probability

Let q be the probability that makes the discounted stock price a martingale. It satisfies

$$100 = \frac{1}{1.02} (120q + 80(1 - q)),$$

so

$$q = \frac{102 - 80}{120 - 80} = 0.55.$$

The option value is therefore

$$V(0) = \frac{1}{1.02} \mathbb{E}^{\mathbb{Q}}[H(1)] = \frac{1}{1.02} (0.55 \cdot 20) = 10.784.$$

► Physical probability:

$$p = 0.70.$$

► Pricing probability:

$$q = 0.55.$$

► Yet

risk-neutral value = replication cost.

Important

- q is not a forecast of the true probability of an up-move.
It is the probability weighting consistent with traded asset prices and absence of arbitrage.

Why the Expected Return Disappears

Under $\mathbb{P}$,

$$\frac{dS(t)}{S(t)} = \mu dt + \sigma dW^{\mathbb{P}}(t).$$

For a replicable derivative $V(t, S)$:

- ▶ construct a delta-hedged portfolio using

$$\Delta = \frac{\partial V}{\partial S};$$

- ▶ the stock-price risk is locally eliminated;
- ▶ the expected stock return μ cancels from the hedged portfolio;
- ▶ absence of arbitrage then gives

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0.$$

Central result

- ▶ *The expected stock return μ matters for forecasting the stock. It does not appear in the Black–Scholes pricing equation for a replicable derivative.*

Learn more

- ▶ For delta hedging, Black–Scholes pricing and Monte Carlo simulation, see the [Computational Finance course, Lecture CF3: Option Pricing and Simulation in Python](#).

Quant Rule #2

For arbitrage-free valuation,

$$V(t) = M(t) \mathbb{E}^{\mathbb{Q}} \left[\frac{H(T)}{M(T)} \mid \mathcal{F}(t) \right].$$

- ▶ Use $\mathbb{P}$ for:
 - ▶ forecasting and expected returns;
 - ▶ historical risk analysis;
 - ▶ real-world scenarios and backtesting.
- ▶ Use $\mathbb{Q}$ for:
 - ▶ arbitrage-free derivative pricing;
 - ▶ calibration to market prices;
 - ▶ valuation consistent with traded assets.

Quant Rule #2

- ▶ *A derivative price is not obtained by simply discounting its expected payoff under an arbitrary probability measure. The probability measure must match the purpose of the calculation.*

Final message

- ▶ *The expected return μ is essential for forecasting the stock. It is not an input to the Black–Scholes price of a replicable derivative.*

Quant Mistake #3: Changing the Numeraire without Changing the Drift

The incorrect belief

- ▶ After changing the numeraire and the associated probability measure, the same stochastic differential equations can be used without modification.

The mistake

- ▶ *Changing the numeraire changes the probability measure. Changing the probability measure generally changes the drift terms of the modeled processes.*
- ▶ The diffusion structure remains unchanged.
- ▶ The Brownian motion changes.
- ▶ Therefore, the drift must change consistently.

Typical implementation failure

- ▶ A pricing formula is rewritten under a stock or forward measure, but the risk factors are still simulated using their dynamics under the original risk-neutral measure.

The consequence

- ▶ The wrong process is treated as a martingale and the resulting derivative price is incorrect.

What Changes When We Change the Numeraire?

Let $N(t) > 0$ be a **traded numeraire** and let $X(t)$ be another **traded asset**. Under the measure $\mathbb{Q}^N$ associated with $N(t)$,

$$\boxed{\frac{X(t)}{N(t)} \text{ is a martingale under } \mathbb{Q}^N \Rightarrow \mathbb{E}^{\mathbb{Q}^N} \left[\frac{X(T)}{N(T)} \mid \mathcal{F}(t) \right] = \frac{X(t)}{N(t)}}.$$

- ▶ Under the money-market numeraire $M(t)$:

$$\frac{X(t)}{M(t)} \text{ is a martingale under } \mathbb{Q}.$$

- ▶ Under a new numeraire $N(t)$:

$$\frac{X(t)}{N(t)} \text{ is a martingale under } \mathbb{Q}^N.$$

- ▶ Neither $X(t)$ nor $N(t)$ must itself be driftless.

Key distinction

- ▶ *Changing the numeraire changes which relative price must be a martingale.*

Learn more

- ▶ For numeraires, martingales and changes of measure, see the [Financial Engineering course, Lecture FE2: Understanding of Filtrations and Measures](#).

The Measure Change is Forced by the Numeraire

Under the risk-neutral measure $\mathbb{Q}$,

$$\frac{dX(t)}{X(t)} = r dt + \sigma_X dW^{\mathbb{Q}}(t), \quad \frac{dN(t)}{N(t)} = r dt + \sigma_N dW^{\mathbb{Q}}(t).$$

- ▶ Apply Itô's formula to

$$Y(t) = \frac{X(t)}{N(t)}.$$

- ▶ Then

$$\frac{dY(t)}{Y(t)} = (\sigma_X - \sigma_N)(dW^{\mathbb{Q}}(t) - \sigma_N dt).$$

- ▶ Under the numeraire measure $\mathbb{Q}^N$, $Y(t) = \frac{X(t)}{N(t)}$ must be a martingale and therefore have zero drift.
- ▶ This forces the definition

$$dW^N(t) = dW^{\mathbb{Q}}(t) - \sigma_N dt.$$

- ▶ Hence

$$\frac{d(X/N)}{X/N} = (\sigma_X - \sigma_N) dW^N(t).$$

Central message

- ▶ *The Brownian motion cannot remain unchanged after a change of numeraire. The shift $dW^N = dW^{\mathbb{Q}} - \sigma_N dt$ is exactly what removes the drift of the numeraire-discounted asset.*

Example: Changing to a Stock Numeraire

- Consider two correlated stocks under $\mathbb{Q}$:

$$\frac{dS_1}{S_1} = r dt + \sigma_1 dW_1^{\mathbb{Q}}, \quad \frac{dS_2}{S_2} = r dt + \sigma_2 dW_2^{\mathbb{Q}},$$

with

$$dW_1^{\mathbb{Q}} dW_2^{\mathbb{Q}} = \rho dt.$$

- Choose $N(t) = S_1(t)$ as the new numeraire.
- Under the stock measure $\mathbb{Q}^{S_1}$,

$$\frac{dS_1}{S_1} = (r + \sigma_1^2) dt + \sigma_1 dW_1^{S_1},$$

and

$$\frac{dS_2}{S_2} = (r + \rho\sigma_1\sigma_2) dt + \sigma_2 dW_2^{S_1}.$$

Important

- The drift of S_2 is no longer r . It contains the correction $\rho\sigma_1\sigma_2$.*

The Martingale Test Exposes the Mistake

- Under the S_1 -measure, the relative price

$$R(t) := \frac{S_2(t)}{S_1(t)}$$

must be a martingale.

- Applying Itô's formula to $R(t)$:

$$\frac{dR(t)}{R(t)} = \sigma_2 dW_2^{S_1}(t) - \sigma_1 dW_1^{S_1}(t),$$

so the drift is zero, as required.

- Suppose instead that we claim to work under the S_1 -measure but incorrectly keep the old drifts:

$$\frac{dS_i}{S_i} \stackrel{\text{wrong}}{=} r dt + \sigma_i dW_i^{S_1}.$$

- Then

$$\frac{dR(t)}{R(t)} \stackrel{\text{wrong}}{=} \left(\sigma_1^2 - \rho \sigma_1 \sigma_2 \right) dt + \sigma_2 dW_2^{S_1}(t) - \sigma_1 dW_1^{S_1}(t).$$

Contradiction

- The process obtained from the incorrect dynamics has non-zero drift. Hence it cannot be the dynamics of $S_2(t)/S_1(t)$ under the S_1 -measure, where this ratio must be a martingale.*

The Wrong Drift Produces the Wrong Price

- Consider the payoff

$$H(T) = S_1(T) \mathbf{1}_{\{S_2(T) > K\}}.$$

- Using S_1 as numeraire,

$$\begin{aligned} V(t_0) &= \mathbb{E}^Q \left[\frac{S_1(T)}{M(T)} \mathbf{1}_{\{S_2(T) > K\}} \right] = \mathbb{E}^{S_1} \left[\frac{M(T) S_1(t_0)}{S_1(T) M(t_0)} \frac{S_1(T)}{M(T)} \mathbf{1}_{\{S_2(T) > K\}} \right] \\ &= S_1(t_0) \mathbb{Q}^{S_1} [S_2(T) > K | \mathcal{F}(t_0)]. \end{aligned}$$

- Take $S_1(t_0) = 100$, $S_2(t_0) = 100$, $K = 110$, $r = 2\%$, $\sigma_1 = 25\%$, $\sigma_2 = 30\%$, $\rho = 0.60$, $T - t_0 = 1$.
- Correct stock-measure drift:

$$r + \rho\sigma_1\sigma_2 = 6.5\%.$$

- Correct price:

$$V(t_0) \approx 40.089.$$

- If the old drift $r = 2\%$ is incorrectly retained:

$$\tilde{V}(t_0) \approx 34.420.$$

- Thus, pricing error $\approx -14.1\%$.

Cause

- The payoff was transformed to the stock measure, but the dynamics were not transformed with it.*

Quant Rule #3

Whenever you change numeraire, verify:

1. the new numeraire $N(t)$;
2. the associated measure $\mathbb{Q}^N$;
3. the transformed Brownian motion;
4. the transformed drift of every simulated risk factor.

Then perform the martingale test:

$$\frac{X(t)}{N(t)} \text{ must have zero drift under } \mathbb{Q}^N.$$

The pricing identity is

$$V(t) = N(t) \mathbb{E}^{\mathbb{Q}^N} \left[\frac{H(T)}{N(T)} \mid \mathcal{F}(t) \right].$$

Quant Rule #3

- Changing the numeraire changes the measure, and changing the measure changes the drift.

Learn more

- For numeraires, Girsanov's theorem and the stock measure, see the *Financial Engineering* course, [Lecture FE2: Understanding of Filtrations and Measures](#).
- For forward measures and measure-dependent forwards drifts, see [Lecture FE11: Market Model and Convexity Adjustments](#).

Quant Mistake #4: Applying Ordinary Calculus to Stochastic Processes

The incorrect belief

- ▶ Since both dt and $dW(t)$ become small over a short interval, all second-order terms may be discarded:

$$(dW(t))^2 \stackrel{\text{wrong}}{=} 0.$$

The mistake

- ▶ A Brownian increment *is of order $\sqrt{dt}$, not dt .*
Therefore,

$$(dW(t))^2$$

is of order dt and contributes to the drift.

Ignoring this term can produce:

- ▶ incorrect transformations of stochastic processes;
- ▶ incorrect drifts and probability distributions;
- ▶ incorrect martingale properties;
- ▶ incorrect pricing PDEs and derivative prices.

Central warning

- ▶ A nonlinear function of a stochastic process cannot generally be differentiated using the ordinary chain rule.

Why Second-Order Terms Survive

- For a continuously differentiable deterministic function $x(t)$,

$$\Delta x = x'(t)\Delta t + o(\Delta t) = O(\Delta t).$$

Hence,

$$(\Delta x)^2 = O((\Delta t)^2),$$

which vanishes after summation over a refining time partition.

- For Brownian motion,

$$\Delta W = W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t),$$

so its increments have stochastic order

$$\Delta W = O_p(\sqrt{\Delta t}), \text{ therefore } (\Delta W)^2 = O_p(\Delta t),$$

so squared Brownian increments **contribute at the same order as the drift term**.

- This leads to the Itô multiplication rules

$$(\mathrm{d}t)^2 = 0, \quad \mathrm{d}t \mathrm{d}W_t = 0, \quad (\mathrm{d}W_t)^2 = \mathrm{d}t.$$

Important

- The identity $(\mathrm{d}W_t)^2 = \mathrm{d}t$ is shorthand for the quadratic-variation limit

$$\sum_i (W_{t_{i+1}} - W_{t_i})^2 \longrightarrow T,$$

as the mesh of a partition of $[0, T]$ tends to zero. It is not an ordinary algebraic identity between infinitesimals.

Itô's Lemma Replaces the Ordinary Chain Rule

Suppose

$$dX(t) = \bar{\mu}(t, X)dt + \bar{\sigma}(t, X)dW(t),$$

and

$$Y(t) = g(t, X(t)).$$

Ordinary calculus would suggest

$$dY \stackrel{\text{wrong}}{=} g_t dt + g_x dX.$$

Itô's lemma gives

$$dY = \left(g_t + \bar{\mu}g_x + \frac{1}{2}\bar{\sigma}^2 g_{xx} \right) dt + \bar{\sigma}g_x dW.$$

- ▶ The extra term is

$$\frac{1}{2}\bar{\sigma}^2 g_{xx} dt, \text{ appears because } (dX)^2 = \bar{\sigma}^2 dt.$$

- ▶ It disappears only for:
 - ▶ deterministic processes, $\bar{\sigma} = 0$;
 - ▶ linear transformations, $g_{xx} = 0$.

Learn more

- ▶ For Itô's lemma and quadratic variation, see the *Computational Finance* course, [Lecture CF2: Stock, Options and Stochastics](#).

A Simple Contradiction: $W^2(t)$

- ▶ Let

$$g(W) = W^2, \quad W(0) = 0.$$

- ▶ Ordinary calculus would give

$$dW^2(t) \stackrel{\text{wrong}}{=} 2W(t)dW(t).$$

- ▶ Since

$$\mathbb{E} \left[\int_0^T W(t) dW(t) \right] = 0,$$

this would imply

$$\mathbb{E}[W^2(T)] \stackrel{\text{wrong}}{=} 0.$$

- ▶ But

$$W(T) \sim \mathcal{N}(0, T), \quad \text{so} \quad \mathbb{E}[W^2(T)] = T.$$

- ▶ Itô's lemma gives the missing term:

$$dW^2(t) = 2W(t)dW(t) + dt.$$

- ▶ Thus,

$$W^2(T) = 2 \int_0^T W(t) dW(t) + T.$$

Lesson

- ▶ *The missing dt term is exactly the contribution of quadratic variation.*

Finance Example: The Logarithm of a Stock Price

- ▶ Suppose

$$\frac{dS(t)}{S(t)} = \mu dt + \sigma dW(t).$$

- ▶ For

$$g(S) = \log S,$$

- ▶ Itô's lemma gives

$$d \log S(t) = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW(t).$$

- ▶ Ordinary calculus would incorrectly give

$$d \log S(t) \stackrel{\text{wrong}}{=} \mu dt + \sigma dW(t),$$

missing

$$-\frac{1}{2} \sigma^2 dt.$$

- ▶ Therefore, the correct solution is

$$S(T) = S(0) \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma W(T) \right].$$

Consequence

- ▶ *Missing the Itô correction does not merely approximate the same model poorly. It produces a different stochastic model.*

Derivative Pricing: The Missing Gamma Term

- ▶ Under $\mathbb{Q}$,

$$dS(t) = rS(t)dt + \sigma S(t)dW^{\mathbb{Q}}(t).$$

- ▶ For a derivative $V(t, S)$, Itô's lemma gives

$$dV = \left(V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} \right) dt + \sigma SV_S dW^{\mathbb{Q}}.$$

- ▶ The second-order term is

$$\frac{1}{2}\sigma^2 S^2 V_{SS} dt.$$

- ▶ Using delta hedging gives the Black–Scholes PDE:

$$V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0.$$

If ordinary calculus is used, the Gamma term disappears.

Immediate warning sign

- ▶ *The resulting incorrect PDE contains no σ .
It would imply that an option price does not depend on volatility.*

Learn more

- ▶ For the Black–Scholes PDE, hedging and pricing, see the *Computational Finance* course, [Lecture CF3: Option Pricing and Simulation in Python](#).

Quant Rule #4

Consider a one-year European call:

$$S = 100, \quad K = 100, \quad T = 1, \quad r = 2\%, \quad \sigma = 20\%.$$

- ▶ Correct Black–Scholes price:

$$C \approx 8.916.$$

- ▶ Ignoring the Itô correction gives

$$\tilde{C} = \left(S - Ke^{-rT} \right)^+ \approx 1.980.$$

- ▶ The missing term removes almost the entire time value of the option.

Quant Rule #4

- ▶ Whenever a nonlinear function of a stochastic process is transformed, use Itô's lemma rather than the ordinary chain rule.

$$dg = \left(g_t + \bar{\mu}g_x + \frac{1}{2}\bar{\sigma}^2g_{xx} \right) dt + \bar{\sigma}g_x dW.$$

Final message

- ▶ *The Itô correction is not a minor mathematical detail. It determines the drift, the distribution and ultimately the derivative price.*

Learn more

- ▶ For stochastic calculus and Itô's lemma, see [Lecture CF2: Stock, Options and Stochastics](#).
- ▶ For Black–Scholes pricing and simulation, see [Lecture CF3: Option Pricing and Simulation in Python](#).

Quant Mistake #5: Treating Implied Volatility as a Forecast of Realized Volatility

The incorrect belief

- ▶ If a one-year option has an implied volatility of 25%, the market is predicting that the stock will realize 25% volatility over the next year.

The mistake

- ▶ *Implied volatility is the volatility parameter that reproduces an observed option price in the Black–Scholes formula.*

It is not, by definition, a real-world forecast of future realized volatility.

The following quantities must be distinguished:

implied volatility $\neq$ realized volatility $\neq$ forecast of realized volatility.

Confusing them can lead to:

- ▶ incorrect trading signals;
- ▶ incorrect volatility forecasts;
- ▶ incorrect interpretation of the volatility surface;
- ▶ incorrect conclusions about whether options are cheap or expensive.

Implied Volatility Is an Inverted Option Price

- ▶ Given an observed option price V_C^{mkt} , implied volatility solves

$$BS(\sigma_{\text{impl}}; S(t), K, r, T) = V_C^{mkt}.$$

- ▶ Since the Black–Scholes call price is increasing in volatility,

$$\frac{\partial BS}{\partial \sigma} = \text{Vega} > 0,$$

the market price can be translated into a Black–Scholes volatility:

$$V_C^{mkt} \xrightarrow{BS^{-1}} \sigma_{\text{impl}}.$$

- ▶ A market may quote an option price as

$$V_C^{mkt} = 9.95,$$

or equivalently as

$$\sigma_{\text{impl}} = 25\%.$$

- ▶ Implied volatility is therefore a convenient coordinate for representing option prices.
- ▶ The inversion does not imply that future realized volatility will equal 25%.

Learn more

- ▶ For Black–Scholes pricing and implied-volatility calculations, see the *Computational Finance* course, [Lecture CF3: Option Pricing and Simulation in Python](#).

Realized Volatility and a Volatility Forecast

- ▶ Suppose the instantaneous volatility is an adapted stochastic process $(\sigma_u)_{u \geq 0}$.
- ▶ The future realized variance over $[t, T]$ is the random variable

$$RV_{t,T} = \frac{1}{T-t} \int_t^T \sigma_u^2 du.$$

- ▶ The corresponding realized volatility is

$$\sigma_{\text{real}}(t, T) = \sqrt{RV_{t,T}}.$$

- ▶ At time T , it is determined by the volatility path that actually occurred.
- ▶ A real-world forecast of future realized variance may be defined as

$$F_{t,T}^{\mathbb{P}} = \mathbb{E}^{\mathbb{P}} [RV_{t,T} | \mathcal{F}_t].$$

- ▶ A volatility forecast may then be quoted as $\hat{\sigma}_{t,T}^{\mathbb{P}} = \sqrt{F_{t,T}^{\mathbb{P}}}$.

Thus:

σ_{impl}	: extracted from option prices,
σ_{real}	: observed from the realised future path,
$F_{t,T}^{\mathbb{P}}$	: forecast of future realised variance under $\mathbb{P}$.

Timing matters

- ▶ *Future realized volatility is a random quantity unless the future volatility path is deterministic. Implied volatility and a real-world forecast are known today, but they are not the same object.*

The Volatility Smile Exposes the Problem

- ▶ For one maturity T , suppose the market gives for $\sigma_{\text{impl}}(K, T)$:

$$\sigma_{\text{impl}}(80, T) = 31\%, \quad \sigma_{\text{impl}}(100, T) = 24\%, \quad \sigma_{\text{impl}}(120, T) = 21\%.$$

- ▶ After the period $[t, T]$ has elapsed, however, the stock has only one realized variance:

$$RV_{t,T}.$$

A simple question

- ▶ *If implied volatility were literally the market's forecast of one future realized volatility, which forecast should we use?*

$$31\%, \quad 24\%, \quad \text{or} \quad 21\%?$$

- ▶ Different strikes have different implied volatilities.
- ▶ Each option emphasizes a different part of the risk-neutral distribution.
- ▶ The volatility smile therefore contains information about:
 - ▶ asymmetry and skew;
 - ▶ downside and tail risk;
 - ▶ jumps;
 - ▶ stochastic volatility.

Conclusion

- ▶ The volatility surface is a representation of option prices, not a single forecast of one future path statistic.

Numerical Example: 25% Implied versus 18% Forecast

- ▶ Consider a one-year at-the-money call:

$$S(t) = 100, \quad K = 100, \quad r = 0.$$

- ▶ Suppose the market price implies

$$\sigma_{\text{impl}} = 25\%.$$

- ▶ Then

$$BS(25\%) \approx 9.948.$$

- ▶ Suppose instead that a statistical model forecasts

$$\sigma_{\text{forecast}} = 18\%.$$

- ▶ Using this number in Black–Scholes gives

$$BS(18\%) \approx 7.171.$$

- ▶ Thus,

$\sigma_{\text{impl}} = 25\%, \quad \sigma_{\text{forecast}} = 18\%$
--

is perfectly possible.

The wrong conclusion

- ▶ *It does not automatically follow that the option is overpriced because*

$$\sigma_{\text{impl}} > \sigma_{\text{forecast}}.$$

Quant Rule #5

Whenever the word “volatility” is used, identify precisely which quantity is meant:

- ▶ implied volatility;
- ▶ instantaneous model volatility;
- ▶ realized volatility;
- ▶ real-world volatility forecast;
- ▶ risk-neutral expected variance.

Quant Rule #5

- ▶ Implied volatility is an option-price quantity, not automatically a forecast of future realized volatility.

Price-implied quantity $\neq$ future realization $\neq$ real-world forecast.

Final message

- ▶ *Implied volatility may contain valuable forecasting information. But whether it predicts future realized volatility is an empirical question—not part of its definition.*

Learn more

- ▶ For option pricing and implied-volatility calculations, see the *Computational Finance* course, [Lecture CF3: Option Pricing and Simulation in Python](#).

Quant Mistake #6: Trusting an Implied Volatility because the Root-Finder Converged

The incorrect belief

- ▶ If a numerical solver returns

$$\sigma_{\text{impl}},$$

then the option has a valid and economically meaningful implied volatility.

The mistake

- ▶ *Solver convergence proves only that a numerical procedure stopped. It does not prove that the implied volatility is correct.*

A root-finder may:

- ▶ search for a root that does not exist;
- ▶ become unstable when Vega is close to zero;
- ▶ stop without accurately reproducing the market price;
- ▶ solve the wrong problem because contract inputs are incorrect;
- ▶ return a numerically precise but economically meaningless number.

Numerical convergence $\neq$ valid implied volatility.

First Question: Does an Implied Volatility Exist?

For a European call,

$$BS_C(\sigma) = V_C^{mkt}$$

has a finite positive solution only if the market price satisfies the no-arbitrage bounds

$$\left(S(t) - Ke^{-r\tau} \right)^+ \leq V_C^{mkt} < S(t).$$

Consider

$$S = 100, \quad K = 100, \quad r = 2\%, \quad \tau = 1.$$

The lower bound is

$$100 - 100e^{-0.02} \approx 1.980.$$

Thus:

- ▶ $V_C^{mkt} = 1.50$ has no implied volatility;
- ▶ $V_C^{mkt} = 100.50$ also has no implied volatility;
- ▶ the correct result is an error, not a volatility number.

Implementation rule

- ▶ *Validate the option price before calling the root-finder.*

Small Vega Makes the Inversion Unstable

- Implied volatility is the inverse of the option-pricing function:

$$BS_C(\sigma_{\text{impl}}(V_C^{\text{mkt}})) = V_C^{\text{mkt}} \iff \sigma_{\text{impl}}(V_C^{\text{mkt}}) = BS_C^{-1}(V_C^{\text{mkt}}).$$

- Locally,

$$\left. \frac{\partial BS_C}{\partial \sigma} \right|_{\sigma=\sigma_{\text{impl}}} \frac{\partial \sigma_{\text{impl}}}{\partial V_C^{\text{mkt}}} = 1, \quad \text{so} \quad \boxed{\frac{\partial \sigma_{\text{impl}}}{\partial V_C^{\text{mkt}}} = \frac{1}{\text{Vega}(\sigma_{\text{impl}})}}.$$

- Hence, for a small change in the observed option price,

$$\boxed{\Delta \sigma_{\text{impl}} \approx \frac{\Delta V_C^{\text{mkt}}}{\text{Vega}(\sigma_{\text{impl}})}}.$$

- Large Vega:

stable implied-volatility inversion.

- Small Vega:

tiny price errors can create large IV errors.

- Vega is often small for:
 - deep ITM options;
 - deep OTM options;
 - very short maturities.

Important

- A quoted implied volatility may look precise even when the underlying option price contains almost no information about volatility.*

Newton–Raphson Can Converge Fast—or Explode

- ▶ Newton–Raphson uses

$$\sigma_{n+1} = \sigma_n - \frac{BS_C(\sigma_n) - V_C^{mkt}}{\text{Vega}(\sigma_n)}.$$

- ▶ Consider a short-dated OTM call:

$$S = 100, \quad K = 120, \quad r = 2\%, \quad \tau = 0.1,$$

with true implied volatility

$$\sigma_{\text{impl}} = 20\%.$$

- ▶ If we start from $\sigma_0 = 10\%$.
- ▶ Vega is extremely small and the first Newton step becomes approximately

$$\sigma_1 \approx 3690.75.$$

- ▶ With a better initial value, $\sigma_0 = 30\%$, the same method converges rapidly toward 20%.

Lesson

- ▶ *A valid implied volatility may exist even when Newton fails.*

Learn more

- ▶ For implied-volatility inversion and Newton–Raphson implementation, see the *Computational Finance* course, [Lecture CF4: Implied Volatility](#).

“Converged” Is Not Enough

- ▶ A solver may stop because

$$|\sigma_{n+1} - \sigma_n| < \varepsilon_\sigma.$$

- ▶ But a small volatility step does not prove that the option price has been matched.
- ▶ Always verify the reconstructed price:

$$\left| BS_C(\sigma_{\text{impl}}) - V_C^{\text{mkt}} \right| < \varepsilon_V.$$

A robust validation should check:

- ▶ convergence of the volatility iteration;
- ▶ small pricing residual;
- ▶ finite and positive implied volatility;
- ▶ sufficiently large Vega;
- ▶ tolerance relative to quote precision and bid–ask spread.

Market precision

- ▶ *For low-Vega options, the bid–ask spread may correspond to a wide range of implied volatilities.*
Reporting many decimal places can create false precision.

Quant Rule #6

Before accepting an implied volatility:

1. validate the contract and market inputs;
2. verify the no-arbitrage price bounds;
3. use a safeguarded solver or volatility bracket;
4. monitor Vega and numerical conditioning;
5. reconstruct the market price from the final volatility.

A robust solver can combine:

Newton speed + bracketing robustness.

Quant Rule #6

- ▶ Never trust an implied volatility simply because the root-finder converged. Validate the existence of the root, the conditioning of the inversion, the inputs and the reconstructed price.

Final message

- ▶ *Do not ask only: "Did the solver converge?"*
Also ask:

"Did we solve the right problem, and is the answer stable?"

Learn more

- ▶ For the BS implied-volatility, Newton–Raphson iteration and its Python implementation, see [Lecture CF4: Implied Volatility](#).

Quant Mistake #7: Using One Black–Scholes Volatility for All Strikes and Maturities

The incorrect belief

- ▶ Once one option has been used to determine a volatility σ , the same value can be used to price every option on the same underlying.

The mistake

- ▶ *A single Black–Scholes volatility implies*

$$\sigma_{\text{impl}}(t; K, T) = \sigma$$

for every strike K and maturity T .

This means a completely flat implied-volatility surface.

In real option markets, implied volatility typically varies with:

- ▶ strike K — producing smile or skew;
- ▶ maturity T — producing a volatility term structure.

The consequence

- ▶ A volatility calibrated to one option may price that option exactly while materially mispricing and mis-hedging other options.

One exact calibration point does not validate one volatility everywhere.

What Does One Constant Volatility Imply?

- Under Black–Scholes,

$$\frac{dS(t)}{S(t)} = rdt + \sigma dW^{\mathbb{Q}}(t),$$

with one constant parameter σ .

- If all market option prices were generated by this model, then

$$\sigma_{\text{impl}}(t; K, T) = \sigma \quad \text{for all } K, T.$$

- Therefore:

- for fixed maturity,

$$K \mapsto \sigma_{\text{impl}}(t; K, T)$$

would be flat;

- for fixed strike,

$$T \mapsto \sigma_{\text{impl}}(t; K, T)$$

would also be flat.

Model implication

- A smile, skew or term structure is direct evidence that one constant Black–Scholes volatility cannot reproduce all quoted option prices simultaneously.

Learn more

- For implied volatility and the deficiencies of the Black–Scholes model, see the *Computational Finance* course, [Lecture CF4: Implied Volatility](#).

A Concrete Volatility Surface

- Consider

$$S(t) = 100, \quad r = q = 0.$$

- Suppose the market implies:

$T - t$	K	$\sigma_{\text{impl}}(t; K, T)$
0.25	80	28%
0.25	100	22%
0.25	120	19%
1.00	80	24%
1.00	100	20%
1.00	120	18%

- The data show:

- a strike effect:

$$\sigma_{\text{impl}}(80, T) > \sigma_{\text{impl}}(120, T);$$

- a maturity effect:

$$\sigma_{\text{impl}}(K, 0.25) \neq \sigma_{\text{impl}}(K, 1.00).$$

Interpretation

- The market is describing a two-dimensional volatility surface

$$(K, T) \mapsto \sigma_{\text{impl}}(t; K, T),$$

not one universal volatility number.

Calibrating One Volatility to One Option

- ▶ Choose the one-year at-the-money call:

$$K = 100, \quad T - t = 1.$$

- ▶ Its market implied volatility is

$$\sigma = 20\%.$$

- ▶ The Black–Scholes model therefore reproduces this option exactly:

$$BS_C(20\%; K = 100, T = 1) = V_C^{mkt}(100, 1).$$

- ▶ But other market options have different implied volatilities:

$$24\%, \quad 22\%, \quad 19\%, \quad 18\%, \dots$$

- ▶ Thus:

- ▶ the selected calibration instrument is priced exactly;
- ▶ other strikes and maturities are not;
- ▶ calibration success at one point says nothing about the fit elsewhere.

The wrong conclusion

- ▶ *“The model perfectly fits one option” does not imply “the model correctly prices the option surface.”*

The Flat Volatility Produces Systematic Mispricing

- Using $\sigma = 20\%$ for every option gives:

$T - t$	K	V_C^{mkt}	$BS_C(20\%) - V_C^{mkt}$
0.25	80	20.2947	-0.2548
0.25	100	4.3862	-0.3984
0.25	120	0.1093	+0.0380
1.00	80	22.0317	-0.8458
1.00	100	7.9656	0.0000
1.00	120	1.5987	+0.5486

- The zero error occurs only at the calibration point: $K = 100$, $T - t = 1$.
- More generally, a first-order change in the option value is

$$dV \approx \Delta dS + \text{Vega } d\sigma + \Theta dt + \rho dr + \dots$$

- In the table above we are not moving the underlying, time or rates. We compare two prices at the same S, K, T, r, q , but with different volatility inputs:

$$dS = 0, \quad dt = 0, \quad dr = 0, \quad dq = 0, \quad d\sigma = \sigma^{flat} - \sigma^{mkt}.$$

Hence the local pricing error is

$$V(S, K, T, r, q, \sigma^{flat}) - V(S, K, T, r, q, \sigma^{mkt}) \approx \text{Vega} (\sigma^{flat} - \sigma^{mkt}).$$

- Therefore, even a modest volatility mismatch can create a material pricing error for high-Vega positions.

The Error Also Affects Hedging

- ▶ Consider

$$S = 100, \quad K = 120, \quad T - t = 1, \quad r = q = 0.$$

- ▶ The market implied volatility is

$$\sigma_{\text{impl}} = 18\%.$$

- ▶ Using the market volatility gives approximately

$$V_C^{mkt} = 1.5987, \quad \Delta_C^{mkt} = 0.1780.$$

- ▶ Using the flat volatility

$$\sigma = 20\%$$

gives

$$V_C^{flat} = 2.1473, \quad \Delta_C^{flat} = 0.2085.$$

- ▶ Hence,

$$\Delta_C^{flat} - \Delta_C^{mkt} \approx 0.0305.$$

- ▶ For one million options, this corresponds to roughly 30,500 additional shares in the hedge.

Consequence

- ▶ *The wrong volatility surface changes not only prices. It also changes Greeks, hedges and portfolio risk.*

Quant Rule #7

Using one volatility may still be an acceptable approximation when:

- ▶ the strike range is narrow;
- ▶ the maturity range is narrow;
- ▶ the local surface is nearly flat;
- ▶ the product has limited smile sensitivity;
- ▶ the approximation error has been measured.

But before using one σ , inspect

$$\sigma_{\text{impl}}(t; K, T)$$

over the relevant strike–maturity region.

Quant Rule #7

- ▶ A single Black–Scholes volatility represents a flat volatility surface. Use it globally only when the market data support that approximation.

Volatility is not just σ . It has coordinates (t, K, T) .

Final message

- ▶ *A volatility fitted to one option is a calibration result for that option. It is not a universal property of the underlying asset.*

Learn more

- ▶ For implied-volatility surfaces see [Lecture CF4: Implied Volatility](#). For stochastic-volatility models and smile dynamics, see [Lecture CF7: Stochastic Volatility Models](#).

Quant Mistake #8: Assuming that Delta Hedging Removes Jump Risk

The incorrect belief

- ▶ If an option portfolio is continuously delta hedged,

$$\Delta(t) = \frac{\partial V(t, S)}{\partial S},$$

then all risk caused by movements of the underlying has been removed.

The mistake

- ▶ *Delta hedging removes the first-order exposure to an infinitesimal continuous movement of the underlying.*

For a finite jump from S to S^+ , let $h_J = S^+ - S$. Then, in general,

$$V(t, S + h_J) - V(t, S) \neq \frac{\partial V}{\partial S}(t, S) h_J.$$

- ▶ Delta is a local hedge ratio.
- ▶ A jump moves the stock across a finite distance **instantly**.
- ▶ There is no opportunity to rebalance during the jump.

The consequence

- ▶ A portfolio can be delta neutral immediately before a jump and still experience a large instantaneous P&L.

Why Delta Hedging Works for Continuous Diffusions

- ▶ Under Black–Scholes, the stock price process satisfies

$$dS(t) = rS(t) dt + \sigma S(t) dW^{\mathbb{Q}}(t).$$

- ▶ For a derivative with value V Itô's lemma gives

$$dV = \left(\frac{\partial V}{\partial t} + rS(t) \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S(t)^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S(t) \frac{\partial V}{\partial S} dW^{\mathbb{Q}}(t).$$

- ▶ Consider the hedged portfolio

$$\Pi(t) = V - \Delta(t)S(t).$$

For fixed hedge ratio $\Delta(t)$ over the infinitesimal interval,

$$\begin{aligned} d\Pi(t) &= dV - \Delta(t) dS(t) \\ &= \left(\frac{\partial V}{\partial t} + rS(t) \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S(t)^2 \frac{\partial^2 V}{\partial S^2} - \Delta(t) rS(t) \right) dt \\ &\quad + \sigma S(t) \left(\frac{\partial V}{\partial S} - \Delta(t) \right) dW^{\mathbb{Q}}(t). \end{aligned}$$

What Happens at a Jump?

- ▶ Suppose the stock jumps from

$$S \text{ to } S^+ = Se^J.$$

- ▶ The finite stock jump is

$$h_J = S^+ - S = S(e^J - 1).$$

- ▶ The option value changes exactly by

$$V(t, S^+) - V(t, S) = V(t, Se^J) - V(t, S).$$

- ▶ Consider a hedged portfolio with Δ shares short:

$$\Pi = V - \Delta S \Rightarrow d\Pi = dV - \Delta dS.$$

- ▶ Across the jump, the portfolio P&L, $R_J(J) = \Delta_J \Pi$, is therefore

$$\begin{aligned} R_J(J) &= V(t, S^+) - V(t, S) - \Delta h_J \\ &= V(t, Se^J) - V(t, S) - \Delta S(e^J - 1). \end{aligned}$$

Key point

- ▶ *During a jump, the hedge cannot be rebalanced. The same pre-jump hedge ratio Δ must absorb the whole finite move.*

Local Delta Versus Jump Hedge Ratio

- ▶ The usual delta is a local derivative:

$$\Delta_{\text{loc}} = \frac{\partial V}{\partial S}(t, S) = \lim_{h \rightarrow 0} \frac{V(t, S + h) - V(t, S)}{h}.$$

- ▶ For one particular finite jump $S \rightarrow Se^J$, the hedge ratio that would exactly hedge that jump is the secant slope

$$\Delta_{\text{jump}}(J) = \frac{V(t, Se^J) - V(t, S)}{S(e^J - 1)}.$$

- ▶ These are generally different:

$$\Delta_{\text{jump}}(J) \neq \Delta_{\text{loc}}.$$

- ▶ They coincide only in the infinitesimal-jump limit:

$$\lim_{J \rightarrow 0} \Delta_{\text{jump}}(J) = \frac{\partial V}{\partial S}(t, S).$$

Interpretation

- ▶ *Delta is the slope of the tangent line. A finite jump requires the slope of a secant line. For a nonlinear payoff, these slopes are not the same.*

Residual Jump Risk After Delta Hedging

- ▶ If the portfolio is delta hedged before the jump, then

$$\Delta = \Delta_{\text{loc}} = \frac{\partial V}{\partial S}(t, S).$$

- ▶ The residual jump P&L is

$$R_J(J) = V(t, Se^J) - V(t, S) - \frac{\partial V}{\partial S}(t, S) S(e^J - 1).$$

- ▶ Let $h_J = S(e^J - 1)$. A Taylor expansion around S gives

$$V(t, S + h_J) = V(t, S) + \frac{\partial V}{\partial S}(t, S)h_J + \frac{1}{2} \frac{\partial^2 V}{\partial S^2}(t, S)h_J^2 + \frac{1}{6} \frac{\partial^3 V}{\partial S^3}(t, S)h_J^3 + \dots$$

- ▶ Subtracting the delta hedge term $\frac{\partial V}{\partial S}(t, S)h_J$ leaves

$$R_J(J) = \frac{1}{2} \frac{\partial^2 V}{\partial S^2}(t, S)(S(e^J - 1))^2 + \frac{1}{6} \frac{\partial^3 V}{\partial S^3}(t, S)(S(e^J - 1))^3 + \dots$$

- ▶ For a small jump,

$$R_J(J) \approx \frac{1}{2} \Gamma(t, S)(S(e^J - 1))^2.$$

Key point

- ▶ *Delta hedging removes the linear part of the move. The leading residual jump exposure is convexity, i.e. Gamma, times the squared jump size.*

Why More Frequent Rebalancing Does Not Solve the Problem

- ▶ Increasing the hedging frequency helps when price paths are continuous.
- ▶ As

$$\Delta t \rightarrow 0,$$

the hedge can track smaller and smaller diffusion movements.

- ▶ But if a jump occurs at time τ ,

$$S(\tau-) \rightarrow S(\tau)$$

instantaneously.

- ▶ The portfolio immediately experiences

$$R_J = V(\tau, S(\tau)) - V(\tau, S(\tau-)) - \Delta(\tau-)[S(\tau) - S(\tau-)].$$

- ▶ Rebalancing faster reduces diffusion discretization risk.
- ▶ It cannot create a trading opportunity inside an instantaneous jump.
- ▶ Even a jump-model Delta generally leaves residual jump risk.

Market incompleteness

- ▶ *With random jump sizes, the stock and cash account generally cannot span all sources of risk.*
Additional instruments may reduce jump exposure, but residual risk typically remains.

Quant Rule #8

For portfolios with material jump or gap exposure:

- ▶ use jump-aware valuation when appropriate;
- ▶ use model-consistent Greeks;
- ▶ calculate finite-jump P&L directly:

$$R_J(J) = V(t, Se^J) - V(t, S) - \Delta S(e^J - 1);$$

- ▶ stress several upward and downward jump scenarios;
- ▶ use additional options when appropriate to reduce convexity and tail exposure;
- ▶ maintain limits or capital for residual unhedgeable risk.

Quant Rule #8

- ▶ Delta hedges infinitesimal continuous price risk.
It does not generally hedge finite jump risk.

Delta neutral $\neq$ jump-risk neutral.

Final message

- ▶ *A hedge based on a local derivative cannot, in general, protect against a finite discontinuous move.*

Learn more

- ▶ For jump processes and jump-diffusion models, see [Lecture CF5: Jump Processes](#).
For models combining stochastic volatility and jumps, see [Lecture CF12: Forward Start Options and Model of Bates](#).

Quant Mistake #9: Adding Jumps without Compensating the Drift

The incorrect belief

- ▶ To extend Black–Scholes with jumps, simply add a jump term:

$$\frac{dS(t)}{S(t-)} \stackrel{\text{wrong}}{=} rdt + \sigma dW^{\mathbb{Q}}(t) + (e^J - 1) dX_P^{\mathbb{Q}}(t).$$

The mistake

- ▶ *The jumps themselves contribute to the expected return. If the continuous drift remains equal to r , the total expected return is generally no longer r .*
- ▶ $X_P^{\mathbb{Q}}(t)$ is a Poisson process with intensity ξ_P .
- ▶ J is the random log-jump size.
- ▶ At a jump,

$$S(t) = S(t-)e^J.$$

- ▶ The proportional jump is therefore

$$\mathbb{E} \left[\frac{\Delta S(t)}{S(t-)} \right] = \mathbb{E} \left[\frac{S(t) - S(t-)}{S(t-)} \right] = \mathbb{E} \left[\frac{S(t-)e^J - S(t-)}{S(t-)} \right] = \mathbb{E} [e^J - 1] \neq 0.$$

Since jumps occur with intensity ξ_P , the jump component contributes an average return of $\xi_P \mathbb{E}[e^J - 1]$ per unit time. **Therefore, the drift of the process needs to “compensate” for this effect.**

The Jumps Contribute to the Drift

- ▶ Consider

$$\frac{dS(t)}{S(t-)} = b dt + \sigma dW^{\mathbb{Q}}(t) + (e^J - 1) dX_P^{\mathbb{Q}}(t).$$

- ▶ Define $\kappa_J := \mathbb{E}^{\mathbb{Q}}[e^J - 1]$.

- ▶ Since

$$\mathbb{E}^{\mathbb{Q}}[dX_P^{\mathbb{Q}}(t)] = \xi_P dt,$$

the expected jump contribution is

$$\mathbb{E}^{\mathbb{Q}}[(e^J - 1)dX_P^{\mathbb{Q}}(t)] = \xi_P \kappa_J dt.$$

- ▶ Therefore, the total expected return is

$$b + \xi_P \kappa_J.$$

- ▶ Under the risk-neutral measure this must equal r .

- ▶ Hence,

$$b = r - \xi_P \kappa_J.$$

The correct dynamics are therefore

$$\frac{dS(t)}{S(t-)} = \left(r - \xi_P \mathbb{E}^{\mathbb{Q}}[e^J - 1] \right) dt + \sigma dW^{\mathbb{Q}}(t) + (e^J - 1) dX_P^{\mathbb{Q}}(t).$$

Do Not Compensate Using the Mean Log Jump

- ▶ At a jump, the stock is multiplied by e^J .
- ▶ Hence the proportional stock jump is

$$\frac{S(t) - S(t-)}{S(t-)} = e^J - 1.$$

- ▶ Therefore the drift correction uses $\xi_p \mathbb{E}^{\mathbb{Q}}[e^J - 1]$, and not $\xi_p \mathbb{E}^{\mathbb{Q}}[J]$.

Normally distributed log jumps

- ▶ If $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$, then

$$\kappa_J := \mathbb{E}^{\mathbb{Q}}[e^J - 1] = \exp\left(\mu_J + \frac{1}{2}\sigma_J^2\right) - 1.$$

- ▶ Even if $\mathbb{E}^{\mathbb{Q}}[J] = 0$, generally

$$\mathbb{E}^{\mathbb{Q}}[e^J - 1] > 0,$$

because the exponential function is convex.

- ▶ Thus $r - \xi_p \mathbb{E}^{\mathbb{Q}}[J]$ is generally the wrong drift correction.

Implementation warning

- ▶ Always distinguish the log-jump J from the proportional stock jump $e^J - 1$.

Learn more: For Poisson processes and jump-diffusion models, see *Computational Finance*, [Lecture CF5: Jump Processes](#).

Numerical Example: Average Downward Jumps

- Consider

$$\begin{aligned} S(0) &= 100, & r &= 2\%, \\ \xi_p &= 0.5, & \mu_J &= -10\%, & \sigma_J &= 20\%. \end{aligned}$$

- The expected proportional jump is

$$\begin{aligned} \kappa_J &= \exp\left(-0.10 + \frac{1}{2}0.20^2\right) - 1 \\ &\approx -0.07688. \end{aligned}$$

- Hence,

$$\xi_p \kappa_J \approx -0.03844.$$

- The correct continuous drift is therefore

$$\begin{aligned} b &= r - \xi_p \kappa_J \\ &= 0.02 - (-0.03844) \\ &= 5.844\%. \end{aligned}$$

Why is the continuous drift above the risk-free rate?

- *Because the negative average jump contribution must be offset.*

$$\underbrace{5.844\%}_{\text{continuous}} + \underbrace{(-3.844\%)}_{\text{average jumps}} = \underbrace{2\%}_{\text{total risk-neutral return}}.$$

The Forward Price Exposes the Error

- ▶ For jump-diffusion dynamics,

$$\frac{dS(t)}{S(t-)} = b dt + \sigma dW^{\mathbb{Q}}(t) + (e^J - 1)dX_P^{\mathbb{Q}}(t),$$

we have

$$\mathbb{E}^{\mathbb{Q}}[S(T)] = S(0) \exp((b + \xi_P \kappa_J)T), \quad \kappa_J = \mathbb{E}^{\mathbb{Q}}[e^J - 1].$$

- ▶ With correct compensation,

$$b = r - \xi_P \kappa_J.$$

- ▶ Hence,

$$\begin{aligned} \mathbb{E}_{correct}^{\mathbb{Q}}[S(1)] &= 100 \exp((0.05844 - 0.03844)) \\ &= 100e^{0.02} \approx 102.020. \end{aligned}$$

- ▶ Suppose instead we incorrectly keep $b = r = 2\%$. Then

$$\begin{aligned} \mathbb{E}_{wrong}^{\mathbb{Q}}[S(1)] &= 100 \exp((0.02 - 0.03844)) \\ &= 100e^{-0.01844} \approx 98.173. \end{aligned}$$

- ▶ Therefore,

$$98.173 \neq 102.020.$$

Immediate diagnostic

- ▶ A model called “risk neutral” no longer reproduces the arbitrage-free forward price. The dynamics are inconsistent with the pricing measure.

Monte Carlo: A Simple Unit Test

- ▶ For a time step Δt , the correct simulation is

$$S_{n+1} = S_n \exp \left[\left(r - \xi_p \kappa_J - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_n + \sum_{k=1}^{N_n} J_{n,k} \right].$$

The common mistake is to omit $-\xi_p \kappa_J$.

- ▶ Before pricing any option:
 - ▶ simulate the stock alone;
 - ▶ calculate

$$\hat{F}(0, T) = \frac{1}{N_{\text{paths}}} \sum_{m=1}^{N_{\text{paths}}} S^{(m)}(T);$$

- ▶ verify

$$\hat{F}(0, T) \approx S(0)e^{rT}.$$

Implementation rule

- ▶ *Do not begin derivative calibration until the simulated underlying reproduces the forward curve.*

Learn more

- ▶ For Monte Carlo simulation techniques, see [Lecture CF9: Monte Carlo Simulation](#).

Quant Rule #9

Whenever jumps are added to an asset-price model:

1. calculate

$$\kappa_J = \mathbb{E}[e^J - 1];$$

2. specify the probability measure;
3. compensate the continuous drift for the expected jump return;
4. verify the forward or martingale condition.

Under $\mathbb{Q}$,

$$b = r - \xi_p \mathbb{E}^{\mathbb{Q}}[e^J - 1].$$

Quant Rule #9

- ▶ When adding jumps, compensate their expected contribution to the return. The continuous drift and the jump component must together produce the required total drift.

$$\text{continuous drift} + \text{expected jump return} = \text{target total return.}$$

Final message

- ▶ *Never add a jump term to an existing diffusion and leave the drift unchanged without checking the martingale condition.*

Learn more

- ▶ For Poisson processes, jump compensation and jump-diffusion models, see [Lecture CF5: Jump Processes.](#)

Quant Mistake #10: Confusing Calibration Quality with Model Validity

The incorrect belief

- ▶ If a model reproduces today's market prices with a very small calibration error, then the model must be correct.

The mistake

- ▶ *Calibration measures fit to a selected data set under a selected objective function. It does not prove that the model has:*
 - ▶ *correct dynamics;*
 - ▶ *stable or identifiable parameters;*
 - ▶ *accurate hedge ratios;*
 - ▶ *reliable out-of-sample prices;*
 - ▶ *realistic tail behaviour;*
 - ▶ *correct prices for products outside the calibration set.*

Good calibration $\neq$ valid model.

The consequence

- ▶ A model may fit vanilla options extremely well and still produce poor hedges, unstable parameters and incorrect exotic prices.

What Calibration Actually Proves

- ▶ Let $\vartheta = (\vartheta_1, \dots, \vartheta_p)$ denote the model parameters.
- ▶ A typical calibration solves

$$\vartheta^* = \arg \min_{\vartheta} \sum_{i=1}^m w_i \left[V_i^{mdl}(\vartheta) - V_i^{mkt} \right]^2.$$

- ▶ The result depends on:
 - ▶ the instruments included;
 - ▶ strikes and maturities;
 - ▶ bid, ask or mid prices;
 - ▶ price versus implied-volatility errors;
 - ▶ weights w_i ;
 - ▶ parameter bounds and regularization;
 - ▶ optimizer and starting values;
 - ▶ numerical pricing accuracy.

What calibration proves

- ▶ *It proves only that the chosen parameter set performs well for the chosen objective and chosen data set.*

Learn more

- ▶ For stochastic-volatility calibration and parameter interpretation, see the *Computational Finance* course, [Lecture CF7: Stochastic Volatility Models](#).

A Small Calibration Error Can Hide Important Problems

- ▶ Suppose a model is calibrated to 100 options.
- ▶ For 98 options, the implied-volatility error is

0.10

volatility percentage points.

- ▶ For two short-dated wing options, the error is

3.00

volatility percentage points.

- ▶ The global RMSE is only

$RMSE \approx 0.436$

volatility percentage points.

This may look acceptable. But the largest errors may occur exactly where a barrier, digital or tail-sensitive product has the greatest exposure.

- ▶ Aggregate error does not show where residuals occur.
- ▶ It does not reveal systematic strike or maturity patterns.
- ▶ It does not tell whether the errors matter for the target portfolio.
- ▶ It should also be compared with bid–ask uncertainty.

Lesson

- ▶ *Never validate a calibration using only one aggregate error number.*

Good Fit Does Not Imply Stable Parameters

- ▶ Let $\mathcal{J}(\vartheta)$ denote the calibration objective function, for example the weighted sum of squared pricing errors.
- ▶ A small value of $\mathcal{J}$ means a good fit to the calibration set, but not necessarily stable parameters.
- ▶ For example, suppose the objective function is

$$\mathcal{J}(\vartheta_1, \vartheta_2) = (\vartheta_1 + \vartheta_2 - 1)^2 + 10^{-4}(\vartheta_1 - \vartheta_2)^2.$$

- ▶ The first term forces approximately

$$\vartheta_1 + \vartheta_2 = 1,$$

while the second term is almost flat.

- ▶ Hence many parameter pairs give almost the same calibration error, for example

$$(0.2, 0.8) \quad \text{and} \quad (0.8, 0.2).$$

- ▶ Calibration prices may be well determined.
- ▶ Individual parameters may be poorly identified.
- ▶ Small changes in market data may cause large parameter moves.
- ▶ Unstable parameters can produce unstable Greeks and exotic prices.

Key point

- ▶ *A very small calibration error does not imply unique, stable or economically meaningful parameters.*

Same Vanilla Prices Can Hide Different Dynamics

- ▶ European options with maturity T depend on the marginal distribution of $S(T)$.
- ▶ Therefore, two models may match the same vanilla surface, meaning

$$S_A(T_i) \stackrel{d}{=} S_B(T_i)$$

at the calibrated maturities T_i .

- ▶ But a **forward-start** payoff depends on how the process moves between dates. For example,

$$H = (S(T_2) - S(T_1))^+.$$

- ▶ Under model A , its price is

$$V_A(0) = \mathbb{E}_A^{\mathbb{Q}} \left[e^{-rT_2} (S_A(T_2) - S_A(T_1))^+ \right].$$

- ▶ Under model B , its price is

$$V_B(0) = \mathbb{E}_B^{\mathbb{Q}} \left[e^{-rT_2} (S_B(T_2) - S_B(T_1))^+ \right].$$

- ▶ Even if the marginal distributions agree at T_1 and T_2 , the joint distributions may differ:

$$(S_A(T_1), S_A(T_2)) \not\stackrel{d}{=} (S_B(T_1), S_B(T_2)).$$

Central lesson

- ▶ *Perfect vanilla calibration does not uniquely determine forward dynamics.*

Validation Must Go Beyond Calibration

After calibration, ask practical questions about how the model will be used.

- ▶ Are the largest pricing errors concentrated in one region of the surface?

$$V_i^{mdl} - V_i^{mkt} \quad \text{by strike and maturity.}$$

- ▶ Do the calibrated parameters remain similar tomorrow?
- ▶ Do we get the same parameters from different starting values?
- ▶ Does the price change materially if we refine the grid, integration settings etc?
- ▶ Does the model price instruments that were not used in calibration?
- ▶ Does the model generate realistic moves of the volatility surface?
- ▶ Do the model Greeks produce reasonable hedge P&L?
- ▶ Does the model behave sensibly for the product we actually want to price: barriers, digitals, jumps, tails or path-dependent claims?

Important

- ▶ *A model can be good enough for vanilla marking and still be unreliable for exotic pricing, hedging or risk management.*

Quant Rule #10

Calibration answers:

“Can this model reproduce the selected market prices?”

Validation asks:

“Is this model reliable for the purpose for which we will use it?”

Quant Rule #10

- ▶ Calibration is an optimization exercise. Model validation is an independent assessment of whether the model is suitable for its intended use.

A close calibration fit does not guarantee:

- ▶ unique or stable parameters,
- ▶ realistic dynamics,
- ▶ reliable exotic prices,
- ▶ accurate hedge ratios

Final message

- ▶ *A model can fit every instrument used to calibrate it and still be wrong for the instrument that matters.*

Learn more

- ▶ For the Heston model and calibration behaviour, see [Lecture CF7: Stochastic Volatility Models](#). For forward-start options and model extensions with jumps, see [Lecture CF12: Forward Start Options and Model of Bates](#).

Quant Mistake #11: Assuming that Stochastic Volatility Solves Every Smile Problem

The incorrect belief

- ▶ Once volatility is made stochastic, the model is flexible enough to reproduce any realistic implied-volatility smile or surface.

The mistake

- ▶ *Stochastic volatility adds a specific mechanism: random volatility. It can generate smiles and skews, but it does not automatically add every mechanism that may be present in market prices.*

A stochastic-volatility model may still be missing:

- ▶ instantaneous price jumps;
- ▶ event-driven gap risk;
- ▶ sufficient control of extreme short-dated wings;
- ▶ enough flexibility to control the entire surface independently.

The consequence

- ▶ If the missing market feature comes from a mechanism that is absent from the model, changing the stochastic-volatility parameters is not the same as adding that mechanism.

What Stochastic Volatility Actually Adds

- ▶ In the Heston model,

$$\frac{dS(t)}{S(t)} = rdt + \sqrt{v(t)} dW_S^Q(t),$$

$$dv(t) = \kappa(\bar{v} - v(t))dt + \gamma\sqrt{v(t)} dW_v^Q(t),$$

$$dW_S^Q(t)dW_v^Q(t) = \rho dt.$$

- ▶ Compared with Black–Scholes, the model can generate:
 - ▶ random and mean-reverting variance;
 - ▶ non-lognormal return distributions;
 - ▶ smile curvature through γ and skew through ρ ;
 - ▶ a non-flat maturity structure.

But

- ▶ *The model still has a particular structure. In particular, the stock process is continuous and one parameter vector*

$$(v_0, \kappa, \bar{v}, \gamma, \rho)$$

has to generate all of these effects together.

Learn more

- ▶ For the Heston model and its implied-volatility behaviour, see the *Computational Finance* course, [Lecture CF7: Stochastic Volatility Models](#).

Different Smile Problems Require Different Mechanisms

The volatility smile is a market observation. The modeling task is to identify the mechanism that explains it.

- ▶ **Random volatility and leverage-generated skew:** stochastic volatility.
- ▶ **Short-dated gap or jump risk:** jump-diffusion models.
- ▶ **Stochastic volatility together with jumps:** models such as Bates.
- ▶ **Very accurate control of today's vanilla surface:** local-volatility or richer surface-calibration approaches.
- ▶ **Stochastic dynamics with stronger surface control:** local-stochastic volatility or other richer models.

Key point

- ▶ *A more complicated stochastic-volatility specification is not automatically the right answer. The extension should introduce the mechanism that is actually missing.*

Example: Short-Dated Skew and Jump Risk

- ▶ In the standard Heston model, the stock path is continuous:

$$S(t-) = S(t).$$

- ▶ Therefore, the model does not contain an instantaneous price jump

$$S(t) = S(t-)e^J.$$

- ▶ This matters when short-dated option prices reflect substantial gap or event risk.
 - ▶ Stochastic volatility can generate skew through ρ .
 - ▶ But it **cannot** create a true discontinuous stock move.
 - ▶ A very steep short-dated downside wing may therefore require a different mechanism.
- ▶ A jump extension introduces precisely such a mechanism:

$$\frac{dS(t)}{S(t-)} = \dots + (e^J - 1) dX_P^Q(t).$$

Central lesson

- ▶ *If the missing market feature is jump or gap risk, increasing volatility-of-volatility is not necessarily the right fix.*

Learn more

- ▶ For jump processes, see [Lecture CF5: Jump Processes](#).

One Volatility Factor Cannot Control Everything Independently

In the Heston model, a small number of parameters controls the behaviour of the entire volatility surface.

- ▶ The correlation ρ influences the direction and strength of skew.
- ▶ The volatility-of-volatility γ influences smile curvature.
- ▶ The mean-reversion parameter κ influences how variance behaviour changes across maturities.

These effects are linked. Changing one parameter generally affects several parts of the surface at the same time.

More stochastic volatility $\neq$ independent control of every smile feature.

Important

- ▶ *If different regions of the market surface require largely independent behaviour, a one-factor stochastic-volatility model may simply be too restrictive.*

Diagnose the Problem Before Extending the Model

When a stochastic-volatility model misses part of the smile, first ask **what kind of failure you are observing**.

- ▶ Is the mismatch concentrated at **short maturities**?
- ▶ Does it appear mainly in the **extreme strikes or wings**?
- ▶ Is the problem associated with a **known event or gap risk**?
- ▶ Does changing one parameter improve one region of the surface but **worsen another**?
- ▶ Does the missing feature require a behaviour that the current dynamics cannot generate at all?

Model-design principle

- ▶ **Do not begin by asking which model to add.**
First identify

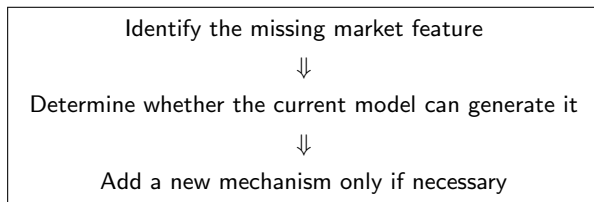
where the model fails and what feature causes the failure.

Practical lesson

- ▶ *A model extension should solve a diagnosed problem, not simply make the model more complicated.*

Quant Rule #11

A stochastic-volatility model should not be expected to reproduce market features generated by mechanisms that are absent from its dynamics.



Quant Rule #11

- ▶ Stochastic volatility is a modeling mechanism, not a universal repair for every smile problem.
Match the model extension to the specific market feature that is missing.

Final message

- ▶ *Diagnose the missing mechanism first; choose the model second.*

Learn more

- ▶ See [Lecture CF7: Stochastic Volatility Models](#), [Lecture CF5: Jump Processes](#), and [Lecture CF12: Forward Start Options and Model of Bates](#).

Quant Mistake #12: Using Fourier Pricing without Checking Numerical Convergence

The incorrect belief

- ▶ If a Fourier pricing routine returns a stable-looking number, the price must be accurate.

The mistake

- ▶ *A Fourier formula may be mathematically exact while its numerical implementation is not.
The computed price depends on truncation, discretization, damping and quadrature choices.*

A plausible output can still contain material numerical error because of:

- ▶ an integration range that is too short;
- ▶ a frequency grid that is too coarse;
- ▶ an unsuitable damping parameter;
- ▶ interpolation error;
- ▶ inaccurate evaluation of the characteristic function.

The consequence

- ▶ Calibration **may fit numerical error instead of model error**, producing biased parameters, unstable Greeks and incorrect exotic prices.

Fourier Pricing Is Still a Numerical Integration Problem

- ▶ A common damped Fourier representation of a European call is

$$C(k) = \frac{e^{-\alpha k}}{\pi} \int_0^{\infty} \operatorname{Re} \left[e^{-iuk} \psi(u) \right] du,$$

where

$$k = \log K$$

and $\psi(u)$ is constructed from the model characteristic function.

- ▶ Numerically, we replace the infinite integral by something like

$$C(k) \approx \frac{e^{-\alpha k}}{\pi} \sum_{j=0}^N w_j \operatorname{Re} \left[e^{-iu_j k} \psi(u_j) \right],$$

with

$$0 \leq u_j \leq U_{\max}.$$

- ▶ $U_{\max}$ controls truncation of the infinite domain.
- ▶ N and the grid spacing control discretization error.
- ▶ α changes the damped integrand and numerical conditioning.

Key distinction

- ▶ *An exact transform formula does not imply an exact numerical price.*

Truncation Error: How Far Is Far Enough?

- ▶ The Fourier integral extends to infinity:

$$\int_0^{\infty} (\dots) du.$$

- ▶ In code, it is truncated:

$$\int_0^{U_{\max}} (\dots) du.$$

- ▶ The omitted tail is the truncation error:

$$\varepsilon_{\text{trunc}} = \int_{U_{\max}}^{\infty} (\dots) du.$$

- ▶ Its size depends on:
 - ▶ the model and its characteristic-function decay;
 - ▶ maturity and model parameters;
 - ▶ strike and payoff;
 - ▶ the damping choice α .

Common mistake

- ▶ *A value of $U_{\max}$ that works for one parameter set need not work after calibration moves the parameters.*

Convergence test

- ▶ Increase $U_{\max}$ until the price is stable to a tolerance that is small relative to the market quote precision and the intended use.

Discretization Error: A Larger Domain Can Make Things Worse

- ▶ For an equally spaced frequency grid,

$$\Delta u = \frac{U_{\max}}{N}.$$

- ▶ Increasing $U_{\max}$ while keeping N fixed makes the grid coarser:

$$U_{\max} \uparrow, \quad N \text{ fixed} \quad \implies \quad \Delta u \uparrow.$$

- ▶ Therefore:

- ▶ increasing $U_{\max}$ reduces truncation error;
- ▶ but it may increase discretization error if N is not increased;
- ▶ the two numerical controls must be tested separately.

- ▶ For FFT implementations, the frequency and log-strike grids are linked. A common relation is

$$\Delta u \Delta k = \frac{2\pi}{N}.$$

Implementation warning

- ▶ *“Use a very large cutoff” is not a convergence strategy. The integration range and grid resolution must be refined together and checked independently.*

Concrete Example: A Price Can Look Converged When It Is Not

- ▶ Consider a one-year Black–Scholes call:

$$S = 100, \quad K = 100, \quad r = 2\%, \quad \sigma = 20\%.$$

- ▶ The analytical benchmark is

$$C_{BS} \approx 8.916037.$$

- ▶ Using a damped Fourier integral with $\alpha = 1.5$ gives, for example,

$U_{\max}$	N	Δu	C_{Fourier}
5	500	0.010	9.307787
10	500	0.020	8.921346
20	500	0.040	8.916017
40	500	0.080	8.916037
160	100	1.600	9.188026

Lesson

- ▶ *Increasing the cutoff from 40 to 160 does not improve the answer when the grid becomes too coarse.*
A large integration range alone is not evidence of convergence.

A Proper Fourier Convergence Test

Do not validate a Fourier implementation with one numerical setup.
Perform several independent checks:

1. Increase $U_{\max}$ while keeping Δu approximately fixed.
2. Refine N at fixed $U_{\max}$.
3. Vary the damping parameter α within its admissible region.
4. Compare alternative quadrature or Fourier formulations.
5. Benchmark against products with known analytical prices.
6. Verify no-arbitrage identities such as put–call parity and the correct forward price.

For two successive numerical settings, monitor

$$\left| V^{(n+1)} - V^{(n)} \right| < \varepsilon_{\text{num}}.$$

Calibration warning

- *Numerical pricing error must be materially smaller than the calibration error you are trying to interpret economically.*

Quant Rule #12

Before trusting a Fourier price, verify:

- ▶ integration-domain convergence;
- ▶ frequency-grid convergence;
- ▶ damping-parameter stability;
- ▶ benchmark and no-arbitrage checks;
- ▶ stability across the parameter region used in calibration.

Quant Rule #12

- ▶ A Fourier formula is only as accurate as its numerical inversion.
Never trust a Fourier price until the numerical parameters have been varied and the result has demonstrated convergence.

Stable-looking output $\neq$ numerically converged price.

Final message

- ▶ *If changing $U_{\max}$, N or the damping parameter changes the price, then the algorithm is still part of the model error.*

Learn more

- ▶ For characteristic functions, Fourier inversion and numerical option pricing, see the *Computational Finance* course, [Lecture CF8: Fourier Transformation for Option Pricing](#).

Quant Mistake #13: Believing that More Monte Carlo Paths Make the Result Correct

The incorrect belief

- ▶ If a Monte Carlo estimate **looks unstable**, simulate more paths.
With enough paths, the result must eventually become correct.

The mistake

- ▶ *More paths reduce sampling error.
They do not remove systematic errors in the quantity being simulated.*

A Monte Carlo result can converge very precisely to the wrong number because of:

- ▶ time-discretization bias;
- ▶ an incorrect stochastic model;
- ▶ wrong drift, payoff or discounting logic;
- ▶ biased path construction or boundary treatment;
- ▶ implementation errors.

More paths $\neq$ more correctness.

What More Paths Actually Do

- ▶ Let $Y_i = e^{-rT} \Pi(s_i(T))$, denote the discounted payoff from a single simulated path.
- ▶ Given N independent simulated paths, the Monte Carlo price estimate is

$$V_{MC}^{(N)} = \frac{1}{N} \sum_{i=1}^N Y_i.$$

- ▶ By the Law of Large Numbers,

$$V_{MC}^{(N)} \longrightarrow \mathbb{E}[Y], \quad N \rightarrow \infty.$$

- ▶ The standard error of the Monte Carlo estimator is

$$\widehat{SE} = \frac{s_Y}{\sqrt{N}}, \quad \text{where } s_Y = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (Y_i - \bar{Y})^2}.$$

- ▶ Therefore:
 - ▶ multiplying N by 4 roughly halves the sampling error;
 - ▶ multiplying N by 100 reduces it by only a factor of 10;
 - ▶ more paths improve precision but do not change the quantity being estimated.

Key distinction

- ▶ Increasing the number of paths reduces random sampling noise around $\mathbb{E}[Y]$.
It does not correct model errors, discretization bias, or coding mistakes.

Four Errors That Must Be Kept Separate

- ▶ A useful decomposition is

$$\widehat{V}_{N,\Delta t} - V^{true} = \underbrace{\widehat{V}_{N,\Delta t} - V_{\Delta t}^{mdl}}_{\text{sampling error}} + \underbrace{V_{\Delta t}^{mdl} - V^{mdl}}_{\text{discretization error}} + \underbrace{V^{mdl} - V^{true}}_{\text{model error}}.$$

- ▶ There may also be implementation error if the algorithm does not compute $\widehat{V}_{N,\Delta t}$ as intended.
 - ▶ $N \uparrow$ reduces only the sampling error.
 - ▶ $\Delta t \downarrow$ addresses time-discretization error.
 - ▶ A better model is needed to reduce model error.
 - ▶ Testing and independent benchmarks are needed to detect coding errors.

Central message

- ▶ *A small Monte Carlo standard error proves only that the simulation has estimated its target precisely. It does not prove that the target itself is correct.*

Concrete Example: Converging to the Wrong Price

- Consider a one-year European call under Black–Scholes:

$$S(0) = 100, \quad K = 100, \quad r = 2\%, \quad \sigma = 20\%.$$

- The exact Black–Scholes dynamics imply

$$S(T) = S(0) \exp\left(\left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T} Z\right), \quad Z \sim \mathcal{N}(0, 1),$$

giving the analytical price

$$C_{BS} \approx 8.9160.$$

- Suppose we do not simulate the exact solution. Instead, we approximate it using a single Euler step:

$$S(T) \approx S^{Euler}(T) = S(0) \left(1 + rT + \sigma\sqrt{T} Z\right).$$

- As the number of paths increases,

$$\hat{C}_N \longrightarrow \mathbb{E}\left[e^{-rT} (S^{Euler}(T) - K)^+\right] \approx 8.8401.$$

- Therefore

$$\underbrace{\hat{C}_N}_{\text{Monte Carlo}} \longrightarrow \underbrace{8.8401}_{\text{Euler-approximated price}} \neq \underbrace{8.9160}_{\text{true Black-Scholes price}}.$$

Why?

- More paths remove sampling noise around the one-step Euler model. They do not remove its discretization bias.

A Tight Confidence Interval Can Be Tightly Wrong

- ▶ For the one-step Euler example, the bias is approximately

$$8.8401 - 8.9160 \approx -0.0759.$$

- ▶ With around one million paths, the Monte Carlo standard error is only about

$$SE \approx 0.012.$$

- ▶ A nominal 95% confidence interval would therefore be roughly

$$8.8401 \pm 1.96 \times 0.012.$$

- ▶ This interval is narrow, but centered on the wrong limiting value.
 - ▶ Confidence intervals quantify sampling uncertainty.
 - ▶ They do not automatically include discretization bias.
 - ▶ They do not include model misspecification or coding errors.

Dangerous conclusion

- ▶ *“The standard error is tiny” does not imply “the price is accurate.”*

A Proper Monte Carlo Convergence Test

A reliable Monte Carlo study varies more than the number of paths.

1. Increase the number of paths

$$N \uparrow$$

to test statistical convergence.

2. Refine the time grid

$$\Delta t \downarrow$$

to test discretization bias.

3. Use analytical or semi-analytical benchmarks whenever available.
4. Check martingale and no-arbitrage identities such as forward reproduction.
5. Compare independent implementations or schemes for difficult products.

Important

- *Variance-reduction methods such as antithetic variables or control variates can reduce sampling variance.*
They do not automatically remove discretization or model bias.

Learn more

- For Monte Carlo simulation, discretization and variance reduction, see [Lecture CF9: Monte Carlo Simulation](#).

Quant Rule #13

Before trusting a Monte Carlo result, ask separately:

- ▶ Has the estimate converged as N increases?
- ▶ Has it converged as the time step is refined?
- ▶ Does the simulation reproduce known moments and forward prices?
- ▶ Does it match analytical benchmarks where available?
- ▶ Are the remaining numerical errors small relative to the required accuracy?

Quant Rule #13

- ▶ More Monte Carlo paths reduce sampling error.
They do not correct a biased discretization, a wrong model or a wrong implementation.

Precision $\neq$ accuracy.

Final message

- ▶ *A Monte Carlo estimate can converge beautifully- to the wrong answer. Always test both statistical convergence and systematic bias.*

Quant Mistake #14: Confusing Weak Convergence with Strong Convergence

The incorrect belief

- ▶ If a discretization scheme gives accurate European option prices as the time step is refined, then the simulated paths **must** also be close to the true stochastic paths.

The mistake

- ▶ *Weak convergence and strong convergence answer different questions. A scheme can reproduce expectations accurately while individual simulated paths remain relatively inaccurate.*
- ▶ Weak convergence concerns distributions and expectations.
- ▶ Strong convergence concerns pathwise closeness under the same source of randomness.
- ▶ The relevant notion depends on what the simulation is used for.

Accurate expectations $\neq$ accurate paths.

Two Different Notions of Convergence

Let $X(T)$ be the exact solution and $X_{\Delta t}(T)$ a numerical approximation.

Weak convergence

- For suitable test functions g ,

$$|\mathbb{E}[g(X_{\Delta t}(T))] - \mathbb{E}[g(X(T))]| \longrightarrow 0.$$

Weak convergence asks whether expectations are reproduced correctly.

Strong convergence

- Using the same underlying Brownian path,

$$\mathbb{E}[|X_{\Delta t}(T) - X(T)|^p]^{1/p} \longrightarrow 0.$$

Strong convergence asks whether the approximate path stays close to the exact path.

Important

- *Never say only that a scheme “converges.”
Specify what converges and in which sense.*

A Simple Counterexample

- ▶ Let

$$X(T) = W(T),$$

and define

$$\tilde{X}(T) = -W(T).$$

- ▶ Since Brownian motion is symmetric,

$$W(T) \stackrel{d}{=} -W(T).$$

- ▶ Therefore, for every integrable payoff g ,

$$\mathbb{E}[g(\tilde{X}(T))] = \mathbb{E}[g(X(T))].$$

- ▶ So the terminal distribution is exactly correct.

- ▶ But pathwise,

$$\begin{aligned} \mathbb{E} \left[|\tilde{X}(T) - X(T)| \right] &= 2\mathbb{E}[|W(T)|] \\ &= 2\sqrt{\frac{2T}{\pi}}, \end{aligned}$$

which is strictly positive.

Lesson

- ▶ *Perfect distributional agreement does not imply pathwise agreement.*

Euler–Maruyama: Different Convergence Rates

- Consider an Itô SDE

$$dX(t) = a(X(t))dt + b(X(t))dW(t).$$

- The Euler–Maruyama scheme is

$$X_{n+1} = X_n + a(X_n)\Delta t + b(X_n)\Delta W_n.$$

- Under standard regularity conditions, Euler–Maruyama typically has:

- strong order 1/2:

$$\text{path error} = O\left((\Delta t)^{1/2}\right);$$

- weak order 1 for sufficiently smooth test functions:

$$\text{expectation error} = O(\Delta t).$$

Interpretation

- *The expectation can converge substantially faster than the individual sample paths.*

Learn more

- For SDE discretization and Monte Carlo convergence, see [Lecture CF9: Monte Carlo Simulation](#).

Which Type of Convergence Matters?

The answer depends on the quantity being computed.

- ▶ European-style pricing

$$V(t) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[h(S(T)) | \mathcal{F}(t)].$$

The target is an expectation, so **weak accuracy** is often the relevant criterion.

- ▶ Path-dependent payoffs depend on the simulated trajectory, for example

$$\max_{0 \leq t \leq T} S(t) \quad \text{or barrier crossings.}$$

Terminal **weak convergence** alone is **not sufficient**.

- ▶ Pathwise hedging, sensitivities and coupled simulations require the numerical path to track the same underlying randomness accurately; **strong convergence can then be essential**.

Common mistake

- ▶ *A scheme validated on European option prices is not automatically validated for barriers, pathwise Greeks or hedging simulations.*

Test Weak and Strong Convergence Separately

Weak-convergence test

- Refine the time step and compare quantities such as

$$\widehat{V}_{\Delta t} = e^{-rT} \frac{1}{N} \sum_{i=1}^N h\left(X_{\Delta t}^{(i)}(T)\right).$$

Check whether

$$\left| \widehat{V}_{\Delta t} - \widehat{V}_{\Delta t/2} \right| = \epsilon_w$$

shrinks consistently.

Strong-convergence test

- Run both simulations using the **same random path** so that differences come only from the time-step size. Measure, for example,

$$\left(\frac{1}{N} \sum_{i=1}^N \left| X_{\Delta t}^{(i)}(T) - X_{\Delta t/2}^{(i)}(T) \right|^2 \right)^{1/2} = \epsilon_s.$$

Implementation rule

- *Weak convergence does not imply strong convergence. Therefore, check prices and path errors separately.*

Quant Rule #14

Before validating a discretization scheme, ask:

- ▶ Is the payoff terminal or path dependent?
- ▶ Are pathwise Greeks, hedging P&L or coupled simulations required?
- ▶ Has the relevant notion of convergence actually been tested?

Quant Rule #14

- ▶ Weak convergence controls expectations and distributions.
Strong convergence controls pathwise closeness under the same source of randomness.
Use the notion of convergence that matches the quantity you are trying to compute.

Correct prices $\nRightarrow$ correct paths.

Final message

- ▶ *Never validate a stochastic simulation scheme using the wrong notion of convergence.*

Learn more

- ▶ For weak and strong convergence of SDE discretizations and their role in Monte Carlo simulation, see [Lecture CF9: Monte Carlo Simulation](#).

Quant Mistake #15: Treating Variance Truncation as a Harmless Fix

The incorrect belief

- ▶ For a CIR or Heston variance process, simulate the variance with a standard Euler step. If the variance becomes negative, simply replace it by zero or apply truncation.

The mistake

- ▶ *The exact square-root variance process is nonnegative, but with the Euler discretization is not. Once truncation, reflection or clipping is introduced, the simulation no longer follows the original finite-step transition law.*

positivity-preserving SDE $\nrightarrow$ positivity-preserving Euler scheme

variance truncation $\neq$ exact CIR transition

The consequence

- ▶ A scheme that avoids numerical failure may cause bias of integrated variance, option prices, Greeks and calibrated parameters.

Continuous-Time Positivity Is Not Discrete-Time Positivity

- ▶ Consider the CIR process

$$dv(t) = \kappa(\bar{v} - v(t))dt + \gamma\sqrt{v(t)}dW(t), \quad v(0) \geq 0.$$

- ▶ The square-root diffusion is designed so that

$$v(t) \geq 0.$$

- ▶ The Feller condition

$$2\kappa\bar{v} \geq \gamma^2$$

is **stronger** than nonnegativity. When the strict inequality holds and $v(0) > 0$, the boundary at zero is unattainable.

- ▶ If the condition is violated, the process may reach zero, but the continuous-time CIR model still **does not** produce negative variance.

Discrete-time warning

- ▶ *The CIR boundary behaviour is a property of the continuous-time process. A finite time-step simulation can violate it unless the numerical scheme is designed to preserve it.*

Naive Euler Can Create Negative Variance

- ▶ The Euler step is

$$\boxed{v_{n+1} = v_n + \kappa(\bar{v} - v_n)\Delta t + \gamma\sqrt{v_n}\sqrt{\Delta t}Z_n}, \quad Z_n \sim \mathcal{N}(0, 1).$$

- ▶ Take

$$v_n = 0.01, \quad \kappa = 3, \quad \bar{v} = 0.04, \quad \gamma = 0.40, \quad \Delta t = \frac{1}{12}.$$

- ▶ Then

$$2\kappa\bar{v} = 0.24 > 0.16 = \gamma^2,$$

so the Feller condition [is satisfied](#).

- ▶ But for only $Z_n = -2$,

$$v_{n+1} \approx 0.01 + 0.0075 - 0.0231 = -0.0056.$$

Key point

- ▶ *The exact process respects the variance state space. The naive Euler approximation does not, even when the Feller condition holds.*

What Truncation Actually Does

- ▶ The naive Euler proposal is

$$\tilde{v}_{n+1} = v_n + \kappa(\bar{v} - v_n)\Delta t + \gamma\sqrt{v_n}\sqrt{\Delta t}Z_n.$$

- ▶ If this proposal is negative, one possible repair is **hard clipping**:

$$v_{n+1} = \max(\tilde{v}_{n+1}, 0).$$

- ▶ Another common choice is **full truncation** Euler. Define

$$v_n^+ := \max(v_n, 0),$$

and simulate

$$v_{n+1} = v_n + \kappa(\bar{v} - v_n^+)\Delta t + \gamma\sqrt{v_n^+}\sqrt{\Delta t}Z_n.$$

- ▶ Different **repairs** change the conditional mean, conditional variance and probability mass near zero.

Key point

- ▶ *These are new discrete-time transition rules. They are finite-step approximations to the CIR dynamics, not the exact CIR transition law.*

Truncation Creates New Probability Mass

- ▶ With the previous parameters, the Euler proposal is

$$\tilde{v}_{n+1} = 0.0175 + 0.01155 Z_n.$$

- ▶ Therefore,

$$\mathbb{Q}[\tilde{v}_{n+1} < 0] \approx 6.5\%.$$

- ▶ With hard clipping, all these outcomes become

$$v_{n+1} = 0.$$

Diagnostic

- ▶ *Here the exact CIR process has an unattainable zero boundary, while the clipped Euler scheme puts about 6.5% probability exactly at zero after one monthly step.*

clipping changes the finite-step distribution

Why the Difference Matters in Heston

- ▶ In Heston,

$$\frac{dS(t)}{S(t)} = rdt + \sqrt{v(t)} dW_S^Q(t).$$

- ▶ The option price depends not only on whether v_n is nonnegative, but on the whole joint simulation of

$$S(t), \quad v(t), \quad \int_0^T v(t)dt.$$

- ▶ A variance scheme can pass the test “no negative square roots” and still **fail** the pricing test.
- ▶ If the variance scheme is poorly behaved:
 - ▶ $\sqrt{v_n}$ may become undefined;
 - ▶ clipping may distort integrated variance;
 - ▶ option prices and implied-volatility smiles may be biased;
 - ▶ calibration **may compensate** for discretization error by changing model parameters.

Central lesson

- ▶ *Fixing a runtime failure is not the same as preserving the intended Heston dynamics.*

Quant Rule #15

Before simulating a square-root variance process:

- ▶ remember that $v(t) \geq 0$ is a property of the continuous-time model;
- ▶ check whether the discretization preserves the same property;
- ▶ specify the exact treatment of negative Euler proposals;
- ▶ treat truncation, reflection and clipping as numerical approximations;
- ▶ test convergence as $\Delta t \downarrow 0$;
- ▶ validate moments, integrated variance, option prices and Greeks.

Quant Rule #15

- ▶ A simulation scheme is part of the model implementation. For CIR and Heston variance, avoiding negative square roots is not enough: the scheme must also be tested for bias and convergence.

valid continuous-time model $\nrightarrow$ valid finite-step simulation.

numerical survival $\neq$ model accuracy.

Final message

- ▶ *Do not repair negative variance after the fact and assume the problem is solved. Judge the scheme by convergence, distributional accuracy and pricing bias.*

Learn more

- ▶ For CIR/Heston simulation and discretization schemes, see [Lecture CF10: Monte Carlo Methods for Stochastic Volatility Models](#).

Quant Mistake #16: Assuming that a Risk-Neutral Discretization Preserves the Martingale Property

The incorrect belief

- ▶ If the stock dynamics are written under the risk-neutral measure with drift r , then any reasonable Monte Carlo discretization automatically preserves the correct forward price.

The mistake

- ▶ *The continuous-time model may be risk-neutral, but the finite time-step simulation is a new discrete-time model. It must be checked separately.*

For each time step, one should have

$$\mathbb{E}_n^{\mathbb{Q}}[S_{n+1}] = S_n e^{r\Delta t}.$$

risk-neutral SDE $\nrightarrow$ martingale-preserving discretization.

The Martingale Test and a Simple Failure

- ▶ In the Heston model,

$$\frac{dS(t)}{S(t)} = r dt + \sqrt{v(t)} dW_S^{\mathbb{Q}}(t).$$

- ▶ Under deterministic rates,

$$\mathbb{E}^{\mathbb{Q}}[S(T)] = S(0)e^{rT},$$

which implies the one-step condition

$$\boxed{\mathbb{E}_n^{\mathbb{Q}}[S_{n+1}] = S_n e^{r\Delta t}.$$

- ▶ A discretization need not preserve this identity exactly. For example, **with a single step size**, with constant variance,

$$S_{n+1} = S_n \left(1 + r\Delta t + \sqrt{v\Delta t} Z_n \right), \quad Z_n \sim N(0, 1),$$

so that

$$\mathbb{E}_n^{\mathbb{Q}}[S_{n+1}] = S_n(1 + r\Delta t) \neq S_n e^{r\Delta t}.$$

Diagnostic

- ▶ *Before trusting option prices, verify that the simulated stock reproduces the forward price. Increasing the number of Monte Carlo paths reduces sampling error, but it does not remove discretization bias.*

The Heston Case: The Martingale Constraint

- ▶ Over one time step, a typical Heston stock update has the form

$$S_{n+1} = S_n e^{r\Delta t} \exp\left(-\frac{1}{2}I_n + X_n\right), \quad I_n \approx \int_{t_n}^{t_{n+1}} v(t) dt,$$

where X_n approximates the stochastic integral driven by $W_S^{\mathbb{Q}}$.

- ▶ Define the one-step stochastic multiplier

$$M_{n+1} := \exp\left(-\frac{1}{2}I_n + X_n\right).$$

Then

$$S_{n+1} = S_n e^{r\Delta t} M_{n+1}.$$

- ▶ The risk-neutral martingale condition requires

$$\mathbb{E}_n^{\mathbb{Q}}[S_{n+1}] = S_n e^{r\Delta t}.$$

- ▶ Hence the discretization must satisfy the one-step identity

$$\boxed{\mathbb{E}_n^{\mathbb{Q}}[M_{n+1}] = \mathbb{E}_n^{\mathbb{Q}}\left[\exp\left(-\frac{1}{2}I_n + X_n\right)\right] = 1.}$$

Where the problem appears

- ▶ *In the exact Heston model this identity holds automatically. In a numerical scheme, I_n , the variance path, and X_n are typically approximated separately. Their joint approximation may therefore break the identity and create a forward-price bias.*

Restoring the Martingale Property

- Suppose the discretization produces the multiplier

$$M_{n+1} = \exp\left(-\frac{1}{2}I_n + X_n\right),$$

but its conditional expectation is not exactly one:

$$m_n := \mathbb{E}_n^{\mathbb{Q}}[M_{n+1}] \neq 1.$$

- The provisional update

$$\tilde{S}_{n+1} = S_n e^{r\Delta t} M_{n+1}$$

then satisfies

$$\mathbb{E}_n^{\mathbb{Q}}[\tilde{S}_{n+1}] = S_n e^{r\Delta t} m_n,$$

and therefore does not preserve the forward price unless $m_n = 1$.

Martingale correction

- Normalize the multiplier before updating the stock:

$$S_{n+1} = S_n e^{r\Delta t} \frac{M_{n+1}}{m_n}, \quad m_n = \mathbb{E}_n^{\mathbb{Q}}[M_{n+1}].$$

This gives

$$\mathbb{E}_n^{\mathbb{Q}}[S_{n+1}] = S_n e^{r\Delta t} \frac{\mathbb{E}_n^{\mathbb{Q}}[M_{n+1}]}{m_n} = S_n e^{r\Delta t}.$$

Important

- This is not a post-processing adjustment of simulated prices. It changes the finite-step transition law and is therefore part of the numerical discretization itself.*

Detecting Martingale Violations

- ▶ Simulate N paths and estimate the forward price

$$\hat{F}(0, T) = \frac{1}{N} \sum_{i=1}^N S_T^{(i)}.$$

- ▶ Compare the estimate with the theoretical value

$$F(0, T) = S_0 e^{rT}.$$

- ▶ Compute a Monte Carlo confidence interval for $\hat{F}(0, T)$.
- ▶ Repeat the test using successively finer time grids

$$\Delta t, \quad \frac{\Delta t}{2}, \quad \frac{\Delta t}{4}.$$

What should happen?

- ▶ For a consistent discretization,

$$\hat{F}(0, T) \longrightarrow S_0 e^{rT}$$

as the time step is refined.

Warning signs

- ▶ *The forward estimate lies consistently outside its confidence interval.*
- ▶ *The bias persists as the number of paths increases.*
- ▶ *The bias decreases only when the time step is reduced.*
- ▶ *Option prices or Greeks change materially under time-step refinement.*

Why Forward-Price Bias Matters

- ▶ The risk-neutral model requires

$$E^{\mathbb{Q}}[S_T] = S_0 e^{rT}.$$

- ▶ If a discretization gives

$$E^{\mathbb{Q}}[S_T] \neq S_0 e^{rT},$$

then it has introduced a drift bias.

- ▶ The simulation is no longer reproducing the intended risk-neutral distribution.
- ▶ The bias can affect:
 - ▶ option prices;
 - ▶ model calibration;
 - ▶ Greeks;
 - ▶ hedging simulations.
- ▶ More Monte Carlo paths reduce sampling error, not discretization bias.

Practical implication

- ▶ *A forward-price error means the simulation is solving a different discrete-time model than the one you intended to price.*

Quant Rule #16

Before trusting a risk-neutral Monte Carlo simulation:

- ▶ verify the martingale condition;
- ▶ compare simulated forwards with $S_0 e^{rT}$;
- ▶ distinguish Monte Carlo noise from discretization bias;
- ▶ check convergence under time-step refinement.

Quant Rule #16

- ▶ A risk-neutral continuous-time model does not automatically produce a risk-neutral discrete-time simulation.

risk-neutral SDE $\nrightarrow$ martingale-consistent discretization.

Final message

- ▶ *Always test whether the simulated stock reproduces the correct forward price. A persistent forward-price bias is evidence that the discretization is not faithfully reproducing the intended model.*

Quant Mistake #17: Computing Monte Carlo Greeks with Independent Random Numbers

The incorrect belief

- ▶ To estimate a Greek by bumping an input, simulate the bumped-up and bumped-down prices **independently**.
With enough paths, the finite difference will become accurate.

The mistake

- ▶ *A Greek is computed from the difference between two nearly identical prices. If the bumped-up and bumped-down valuations use different random numbers, each price contains different simulation noise. When the prices are subtracted, the noise does not cancel well and can hide the true effect of the bump.*

For Delta,

$$\hat{\Delta} = \frac{\hat{V}(S + h; Z) - \hat{V}(S - h; \tilde{Z})}{2h}, \quad Z \neq \tilde{Z}.$$

same bump + different random numbers = noisy Greek

Common Random Numbers Create Pathwise Cancellation

- ▶ Suppose a path is generated from random input

$$Z^{(i)}.$$

- ▶ For both bumped simulations, use the same random input:

$$S_T^{+, (i)} = F(S + h, Z^{(i)}),$$

and

$$S_T^{-, (i)} = F(S - h, Z^{(i)}).$$

- ▶ Then estimate Delta path by path:

$$\hat{\Delta} = \frac{1}{N} \sum_{i=1}^N \frac{H(S_T^{+, (i)}) - H(S_T^{-, (i)})}{2h}.$$

- ▶ The same favorable shock affects both bumped paths.
- ▶ The same unfavorable shock affects both bumped paths.
- ▶ Much of the common payoff noise cancels in the difference.
- ▶ The remaining difference primarily reflects the parameter bump.

Interpretation

- ▶ *For finite-difference Greeks, correlation between simulations is often desirable.*

Why the Difference of Two Prices Can Be Very Noisy

- ▶ Let

$$\widehat{V}_+ := \widehat{V}(S + h), \quad \widehat{V}_- := \widehat{V}(S - h).$$

- ▶ Then

$$\widehat{\Delta} = \frac{\widehat{V}_+ - \widehat{V}_-}{2h}.$$

- ▶ Its variance is

$$\text{Var}(\widehat{\Delta}) = \frac{\text{Var}(\widehat{V}_+) + \text{Var}(\widehat{V}_-) - 2 \text{Cov}(\widehat{V}_+, \widehat{V}_-)}{4h^2}.$$

- ▶ With independent simulations,

$$\text{Cov}(\widehat{V}_+, \widehat{V}_-) = 0.$$

- ▶ With common random numbers, the two bumped payoffs are usually strongly positively correlated:

$$\text{Cov}(\widehat{V}_+, \widehat{V}_-) > 0.$$

Key point

- ▶ *Positive correlation between the bumped valuations reduces the variance of their difference.*

Concrete Example: Same Delta, Very Different Noise

- ▶ Consider a one-year Black–Scholes call:

$$S = 100, \quad K = 100, \quad r = 2\%, \quad \sigma = 20\%.$$

- ▶ Use a central bump $h = 0.10$, and $N = 100,000$, paths for each bumped valuation.
- ▶ The analytical Delta is approximately

$$\Delta_{BS} \approx 0.5793.$$

- ▶ The standard error measures how much the estimated Delta would typically vary if the simulation were repeated:

$$SE(\hat{\Delta}) = \sqrt{\text{Var}(\hat{\Delta})}.$$

- ▶ For exact GBM simulation, the standard error of the finite-difference Delta is approximately:

simulation design	standard error of $\hat{\Delta}$
independent random numbers	0.31
common random numbers	0.0019

Result

- ▶ *The model, bump size and path count are unchanged.
Only the coupling of the random numbers changes- yet the Greek becomes orders of magnitude more precise.*

Small Bumps Amplify Monte Carlo Noise

- ▶ A finite-difference Delta estimator is

$$\hat{\Delta} = \frac{\hat{V}(S+h) - \hat{V}(S-h)}{2h}.$$

- ▶ Each Monte Carlo price contains sampling error:

$$\hat{V}(S+h) = V(S+h) + \varepsilon_+,$$

$$\hat{V}(S-h) = V(S-h) + \varepsilon_-.$$

- ▶ Substituting into the Delta estimator gives

$$\hat{\Delta} = \frac{V(S+h) + \varepsilon_+ - (V(S-h) + \varepsilon_-)}{2h}.$$

- ▶ Rearranging,

$$\hat{\Delta} = \frac{V(S+h) - V(S-h)}{2h} + \frac{\varepsilon_+ - \varepsilon_-}{2h}.$$

Key observation

- ▶ *The Greek estimate contains both the finite-difference signal and the difference of the Monte Carlo pricing errors.*

Smaller Bumps Can Make Independent Simulations Worse

- Recall the decomposition

$$\hat{\Delta} = \frac{V(S+h) - V(S-h)}{2h} + \frac{\varepsilon_+ - \varepsilon_-}{2h}.$$

- The first term is the finite-difference approximation to Delta.
- The second term is the contribution from Monte Carlo pricing errors.
- Reducing the bump $h \downarrow 0$ usually improves the first term by reducing finite-difference bias.
- However, the same reduction increases $\frac{1}{2h}$, so the error component

$$\frac{\varepsilon_+ - \varepsilon_-}{2h} \text{ is magnified.}$$

- With independent simulations,

$$\varepsilon_+ \text{ and } \varepsilon_-$$

are unrelated, so their difference can be substantial.

- With common random numbers,

$$\varepsilon_+ - \varepsilon_- \approx 0,$$

and much of the error cancels before division by $2h$.

Bias-variance tradeoff

- Large h : more finite-difference bias.* *Small h : more amplified Monte Carlo noise.*

Common Random Numbers Help—but Do Not Solve Everything

Common random numbers are a variance-reduction technique.

They do not remove every source of Greek error.

A robust Monte Carlo Greek calculation should also test:

- ▶ bump-size convergence;
- ▶ time-discretization bias;
- ▶ sufficient path count;
- ▶ stability for discontinuous or path-dependent payoffs;
- ▶ alternative estimators when appropriate.

Other Greek estimators may include:

pathwise differentiation, likelihood-ratio methods,

or algorithmic and adjoint differentiation when applicable.

Important

- ▶ *Common random numbers reduce variance.*
They do not remove finite-difference bias or repair an inappropriate Greek estimator.

Learn more

- ▶ For Monte Carlo Greeks and variance-reduction techniques, see [Lecture CF11: Hedging and Monte Carlo Sensitivites](#).

Quant Rule #17

When computing Greeks by bump-and-revalue Monte Carlo:

- ▶ reuse the same random numbers across bumped valuations;
- ▶ preserve the same Brownian paths, jump draws and other random inputs;
- ▶ study sensitivity to the bump size h ;
- ▶ report the standard error of the Greek, not only of the prices;
- ▶ benchmark against analytical Greeks where available.

Quant Rule #17

- ▶ For finite-difference Monte Carlo Greeks, use common random numbers so that the bumped valuations move together and common simulation noise cancels.

Final message

- ▶ *Independent random numbers can turn two individually accurate price estimates into a very poor noise cancellation estimate.*

Quant Mistake #18: Assuming that Delta Hedging Removes All Financial Risk

The incorrect belief

- ▶ If an option portfolio is delta neutral,

$$\Delta_{portfolio} = 0,$$

then the position is essentially risk free.

The mistake

- ▶ *Delta measures only the local first-order sensitivity to the underlying price. A portfolio can have zero Delta and still carry substantial exposure to other risk factors and to nonlinear spot moves.*

A delta-neutral portfolio may still have:

- ▶ Gamma and discrete-rebalancing risk;
- ▶ Vega and volatility-surface risk;
- ▶ jump and gap risk;
- ▶ model and parameter risk;
- ▶ transaction-cost, liquidity and funding risk.

Delta neutral $\neq$ risk free.

When Can Delta Hedging Replicate an Option?

- ▶ In the Black–Scholes model,

$$\frac{dS(t)}{S(t)} = rdt + \sigma dW^{\mathbb{Q}}(t),$$

and a derivative value is

$$V(t, S).$$

- ▶ Choosing

$$\Delta(t) = \frac{\partial V}{\partial S}$$

eliminates the instantaneous Brownian term in the hedged portfolio.

- ▶ Perfect replication additionally relies on strong assumptions:
 - ▶ continuous stock paths;
 - ▶ continuous, frictionless rebalancing;
 - ▶ the correct model and parameters;
 - ▶ no transaction costs or liquidity constraints;
 - ▶ borrowing and lending at the modeled funding rate.

Important distinction

- ▶ *Under the idealized model, dynamic Delta hedging can replicate the option. In practice, “Delta neutral now” is not the same statement as “all future risk has been removed.”*

Discrete Hedging Leaves Residual Gamma Risk

- ▶ A continuously rebalanced Black–Scholes hedge is an idealization. Real hedges are adjusted at discrete times.
- ▶ To isolate the effect of discrete spot hedging, assume volatility is unchanged over the hedge interval: $d\sigma \approx 0$.
- ▶ For a small move in spot and time,

$$dV \approx \Delta dS + \frac{1}{2}\Gamma(dS)^2 + \Theta dt.$$

- ▶ Construct a delta-hedged portfolio:

$$\Pi = V - \Delta S.$$

- ▶ Portfolio P&L:

$$d\Pi = dV - \Delta dS.$$

- ▶ Substitute the option expansion:

$$d\Pi = \left(\Delta dS + \frac{1}{2}\Gamma(dS)^2 + \Theta dt \right) - \Delta dS.$$

- ▶ The first-order spot exposure cancels:

$$d\Pi \approx \frac{1}{2}\Gamma(dS)^2 + \Theta dt$$

Interpretation

- ▶ *For unchanged volatility, delta hedging removes the linear spot term. The remaining short-interval P&L comes from curvature, through Gamma, and time decay, through Theta.*

Delta Removes Only the Linear Spot Exposure

- For small changes, an option-value expansion is

$$\Delta V \approx \Delta dS + \frac{1}{2}\Gamma(dS)^2 + \text{Vega } d\sigma + \Theta dt + \dots$$

- After Delta hedging,

$$\Delta_{\text{portfolio}} = 0,$$

but the remaining P&L still contains terms such as

$$\frac{1}{2}\Gamma(dS)^2 + \text{Vega } d\sigma + \Theta dt + \dots$$

- For example, if

$$\Gamma = 0.02 \quad \text{and} \quad |dS| = 5,$$

then the second-order spot contribution is approximately

$$\frac{1}{2}(0.02)(5)^2 = 0.25$$

per option, even though the initial Delta exposure is zero.

Key point

- *Delta removes the tangent-line exposure, not the curvature of the option.*

Delta Neutral Does Not Mean Vega Neutral

- ▶ The previous slide isolated spot moves by assuming $d\sigma \approx 0$.
- ▶ In practice, implied volatility also moves. The option-value expansion therefore contains a separate Vega term:

$$dV \approx \Delta dS + \frac{1}{2} \Gamma (dS)^2 + \text{Vega } d\sigma + \Theta dt + \dots$$

- ▶ Delta neutrality means $\Delta_{\text{portfolio}} = 0$, so the first-order spot term is removed.
- ▶ But the volatility term remains:

$$\Delta V_{\text{vol}} \approx \text{Vega}_{\text{portfolio}} \Delta\sigma.$$

- ▶ Suppose

$$\text{Vega}_{\text{portfolio}} = 50,000$$

per volatility point.

- ▶ A volatility move of

$$\Delta\sigma = 2$$

volatility points produces approximately

$$\Delta V_{\text{vol}} \approx 50,000 \times 2 = 100,000.$$

The portfolio had zero Delta before the volatility move, but it was not Vega neutral.

- ▶ implied volatility changes can generate P&L independently of Delta;
- ▶ smile and skew changes create additional volatility-surface risk;

Lesson

- ▶ *Delta hedging manages first-order spot exposure. It does not hedge implied-volatility exposure. Vega must be measured and managed separately.*

Some Risks Cannot Be Removed by Faster Delta Hedging

Increasing hedge frequency may reduce diffusion rebalancing error, but it does not remove every source of risk.

- ▶ **Jump risk** there is no opportunity to rebalance inside an instantaneous gap.
- ▶ **Model risk**: the hedge ratios are only as good as the model generating them.
- ▶ **Transaction costs and liquidity**: frequent trading can itself become expensive or impossible.
- ▶ **Funding and collateral risk**: real financing rates and margin requirements may differ from model assumptions.
- ▶ **Parameter risk**: volatility, correlation and other inputs move through time.

Practical principle

- ▶ *Delta hedging is one layer of risk management.
It is not a substitute for stress testing, Greek limits, model validation and liquidity management.*

Learn more

- ▶ For Monte Carlo Greeks and hedging analysis, see [Lecture CF11: Hedging and Monte Carlo Sensitivites](#).

Quant Rule #18

Before calling a portfolio “hedged,” check more than Delta:

- ▶ Gamma and nonlinear spot scenarios;
- ▶ Vega, smile and term-structure exposure;
- ▶ jump and gap scenarios;
- ▶ model and parameter sensitivity;
- ▶ transaction costs, liquidity and funding constraints.

Quant Rule #18

- ▶ Delta hedging removes local first-order spot exposure.
It does not remove all financial risk unless the assumptions of the replication model actually hold.

Delta hedging = risk reduction, not universal risk elimination.
--

Final message

- ▶ *A portfolio can be Delta neutral and still lose money from curvature, volatility, jumps, model error, trading costs or funding.*

Quant Mistake #19: Assuming Today's Vanilla Surface Determines Forward-Start Prices

The incorrect belief

- ▶ Once a model is calibrated to today's vanilla implied-volatility surface, the price of a forward-start option is essentially determined.

The mistake

- ▶ *Today's vanilla option prices constrain the distributions of the underlying at individual maturities.*
A forward-start option also depends on how the market evolves between the reset date T_1 and the maturity T_2 .

For a proportional strike k , the payoff is

$$(S(T_2) - kS(T_1))^+.$$

Therefore:

- ▶ the strike is determined by the future level $S(T_1)$;
- ▶ the relevant option period is $[T_1, T_2]$;
- ▶ pricing depends on future conditional market dynamics.

spot vanilla calibration $\nrightarrow$ unique forward-start price.

The Payoff Reveals the Missing Information

- ▶ Let

$$t_0 < T_1 < T_2.$$

- ▶ A standard European call has payoff

$$(S(T_2) - K)^+,$$

where K is known at t_0 .

- ▶ A forward-start call has payoff

$$(S(T_2) - kS(T_1))^+ = S(T_1) \left(\frac{S(T_2)}{S(T_1)} - k \right)^+.$$

- ▶ Thus, the payoff depends on:
 - ▶ the future reset level $S(T_1)$;
 - ▶ the return from T_1 to T_2 ;
 - ▶ the conditional distribution of that future return.

Key point

- ▶ *Today's vanilla surface describes option prices starting today. A forward-start option is a claim on an option period that begins only at the future reset date T_1 .*

Pricing by Conditional Expectation

- Under the pricing measure $\mathbb{Q}$,

$$V(t_0) = e^{-r(T_2-t_0)} \mathbb{E}^{\mathbb{Q}} [(S(T_2) - kS(T_1))^+ \mid \mathcal{F}(t_0)] .$$

- Splitting the discount factor at T_1 and applying the tower property,

$$\begin{aligned} V(t_0) &= e^{-r(T_1-t_0)} \mathbb{E}^{\mathbb{Q}} \left[\mathbb{E}^{\mathbb{Q}} \left[e^{-r(T_2-T_1)} (S(T_2) - kS(T_1))^+ \mid \mathcal{F}(T_1) \right] \mid \mathcal{F}(t_0) \right] \\ &= e^{-r(T_1-t_0)} \mathbb{E}^{\mathbb{Q}} [V(T_1) \mid \mathcal{F}(t_0)] , \end{aligned}$$

where

$$V(T_1) := e^{-r(T_2-T_1)} \mathbb{E}^{\mathbb{Q}} [(S(T_2) - kS(T_1))^+ \mid \mathcal{F}(T_1)] .$$

Interpretation

- The time- t_0 value is the discounted expected value of the future option received at the reset date T_1 .

Learn more

- For filtrations, martingales, etc., see [Lecture FE2: Understanding of Filtrations and Measures](#).

What Determines the Value at the Reset Date?

- ▶ At T_1 , the reset level $S(T_1)$ is known. Therefore,

$$\begin{aligned} V(T_1) &= e^{-r(T_2-T_1)} \mathbb{E}^{\mathbb{Q}} \left[(S(T_2) - kS(T_1))^+ \mid \mathcal{F}(T_1) \right] \\ &= S(T_1) e^{-r(T_2-T_1)} \mathbb{E}^{\mathbb{Q}} \left[\left(\frac{S(T_2)}{S(T_1)} - k \right)^+ \mid \mathcal{F}(T_1) \right]. \end{aligned}$$

Thus:

- ▶ $S(T_1)$ scales the value of the future option;
- ▶ the option's moneyness depends on the return $S(T_2)/S(T_1)$;
- ▶ the conditional distribution of this return depends on the market state at T_1 ;
- ▶ in a stochastic-volatility model, $v(T_1)$ affects the future variance and therefore the future return distribution.

Today's vanilla surface constrains marginal distributions at individual maturities, but does not generally determine these future conditional dynamics.

Central lesson

- ▶ *Forward-start pricing tests the model's forward dynamics, not only its fit to today's vanilla prices.*

Black–Scholes Is a Special Simplification

- ▶ Under Black–Scholes, $S(t) = S(0) \exp\left[\left(r - \frac{1}{2}\sigma^2\right)t + \sigma W(t)\right]$. so:

$$\frac{S(T_2)}{S(T_1)} = \exp\left[\left(r - \frac{1}{2}\sigma^2\right)(T_2 - T_1) + \sigma(W(T_2) - W(T_1))\right].$$

- ▶ Since the Brownian increment $W(T_2) - W(T_1) \sim N(0, T_2 - T_1)$ is independent of $\mathcal{F}(T_1)$, the ratio $\frac{S(T_2)}{S(T_1)}$ is independent of $S(T_1)$ and depends only on $\tau := T_2 - T_1$.
- ▶ Because the call payoff is homogeneous in spot and strike, the forward-start call simplifies to

$$V(t_0) = S(t_0) \left[\Phi(d_1) - ke^{-r\tau} \Phi(d_2) \right],$$

where

$$d_1 = \frac{\log(1/k) + \left(r + \frac{1}{2}\sigma^2\right)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Do not overgeneralize

- ▶ *This simplification follows from Black–Scholes scale invariance, independent lognormal increments and constant volatility. It is not a generic property of stochastic-volatility or jump models.*

Numerical Example: Using the Wrong Volatility

- ▶ Consider

$$S(0) = 100, \quad r = 2\%, \quad T_1 = 1, \quad T_2 = 2, \quad k = 1.$$

- ▶ Suppose that the instantaneous volatility is

$$\sigma_1 = 15\% \text{ on } [0, T_1], \quad \sigma_2 = 25\% \text{ on } [T_1, T_2].$$

- ▶ The two-year ATM implied volatility is

$$\sigma_{\text{imp}}(0, T_2) = \sqrt{\frac{1}{T_2} \int_0^{T_2} \sigma^2(t) dt} = \sqrt{\frac{\sigma_1^2 T_1 + \sigma_2^2 (T_2 - T_1)}{T_2}} \approx 20.62\%.$$

- ▶ The correct forward-start value uses the volatility of the future period,

$$\sigma_{\text{fwd}}(T_1, T_2) = 25\%, \text{ and is } V_{\text{correct}} \approx 10.8706.$$

- ▶ Using today's two-year implied volatility instead gives

$$V_{\text{wrong}} \approx 9.1567,$$

an underpricing of approximately

$$\frac{10.8706 - 9.1567}{10.8706} \approx 15.8\%.$$

Why the difference?

- ▶ *The two-year implied volatility averages variance over $[0, T_2]$.
The forward-start option is driven by uncertainty over $[T_1, T_2]$.*

Beyond Black–Scholes: The Forward Smile Matters

- ▶ In a stochastic-volatility model, the variance level $v(T_1)$ observed at the reset date affects the future variance over $[T_1, T_2]$.
- ▶ It therefore affects the conditional distribution of $\frac{S(T_2)}{S(T_1)}$ and hence the value of

$$S(T_1) \left(\frac{S(T_2)}{S(T_1)} - k \right)^+.$$

- ▶ Thus:
 - ▶ $S(T_1)$ scales the forward-start payoff;
 - ▶ $v(T_1)$ influences the future return distribution;
 - ▶ different volatility dynamics produce different forward smiles.
- ▶ Therefore, forward-start options are sensitive to:
 - ▶ forward implied volatility;
 - ▶ future smile and skew dynamics;
 - ▶ the joint evolution of spot and volatility.

Common mistake

- ▶ *Using today's T_2 vanilla implied volatility as if it were the volatility for the future period $[T_1, T_2]$ is generally unjustified.*

Learn more

- ▶ For forward-start options, forward implied volatility and the Bates model, see [Lecture CF12: Forward Start Options and Model of Bates](#).

Quant Rule #19

Before pricing a forward-start option:

- ▶ identify the reset date T_1 and maturity T_2 ;
- ▶ distinguish today's vanilla smile from the future forward smile;
- ▶ identify the model assumptions governing the period $[T_1, T_2]$;
- ▶ test sensitivity to forward-volatility and joint-dynamics assumptions.

Quant Rule #19

- ▶ Today's vanilla surface does not, by itself, determine the price of a forward-start option.
The price depends on the model's future conditional dynamics and forward smile.

spot-smile calibration $\neq$ forward-smile calibration.

Final message

- ▶ *Never substitute today's maturity volatility for the volatility of a future option period without verifying the model assumptions that justify the substitution.*

Quant Mistake #20: Ignoring the Path when Pricing Path-Dependent Derivatives

The incorrect belief

- ▶ A payoff maturing at T can be valued using only the risk-neutral distribution of $S(T)$.

Hence, if two models satisfy

$$S_1(T) \stackrel{d}{=} S_2(T),$$

they should assign the same price to any derivative maturing at T .

The mistake

- ▶ *This is true only for payoffs of the form $f(S(T))$. Path-dependent payoffs depend on the trajectory*

$$\{S(t) : 0 \leq t \leq T\},$$

so their prices generally depend on more than the terminal distribution.

Examples include:

- ▶ barrier and lookback options;
- ▶ Asian options;
- ▶ cliquets and autocallables; etc.

same terminal distribution $\neq$ same path-dependent price.

Terminal Information Is Not Enough

- ▶ A European payoff has the form

$$H^{vanilla} = h(S(T)).$$

- ▶ Its value depends on the terminal distribution of $S(T)$:

$$V(0) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[h(S(T))].$$

- ▶ A path-dependent payoff has the form

$$H^{path} = F(\{S(t) : 0 \leq t \leq T\}).$$

- ▶ It may depend on quantities such as

$$\max_{0 \leq t \leq T} S(t), \quad \min_{0 \leq t \leq T} S(t), \quad \frac{1}{T} \int_0^T S(t) dt.$$

Central point

- ▶ *Matching the marginal law of $S(T)$ does not determine the joint law of the entire path.*

Concrete Example: Same Terminal Price, Different Barrier Payoff

- Consider a down-and-out call $(S(T) - K)^+ \mathbf{1}_{\{\min_{0 \leq t \leq T} S(t) > B\}}$, with

$$K = 100, \quad B = 80,$$

and terminal stock price

$$S(T) = 110.$$

- Compare two possible paths:

	minimum along the path	payoff
Path A	90	10
Path B	75	0

- Both paths finish at the same value:

$$S(T) = 110.$$

- But Path B crossed the barrier and knocked the option out.

Lesson

- The terminal stock price alone cannot determine the payoff. The barrier event is encoded in the path.*

Discrete Monitoring Can Miss the Path

- Suppose a continuously monitored barrier is checked only on a simulation grid

$$0 = t_0 < t_1 < \dots < t_N = T.$$

- Even if

$$S(t_i) > B \quad \text{and} \quad S(t_{i+1}) > B,$$

the continuous path may still cross B between the two dates.

- For a lognormal diffusion with constant volatility, a Brownian-bridge correction can quantify this missed crossing probability:

$$p_{hit} = \exp \left(- \frac{2 \log(S(t_i)/B) \log(S(t_{i+1})/B)}{\sigma^2 \Delta t} \right).$$

Implementation warning

- *No barrier breach on the simulation grid does not mean no barrier breach along the continuous path.*
*Even when both simulated endpoints lie above the barrier, there is a **non-zero conditional probability** that the barrier was crossed between them.*

Asian Options Need the Running Path State

- ▶ For an arithmetic-average Asian call,

$$H = \left(\frac{1}{m} \sum_{j=1}^m S(t_j) - K \right)^+.$$

- ▶ Knowing only

$$S(T)$$

is insufficient.

- ▶ For example, with $K = 100$ and the same terminal value $S(T) = 110$:

	observed path values	average payoff
Path A	100, 105, 110	5
Path B	100, 80, 110	0

- ▶ The pricing state must therefore include the relevant path statistic, for example the running sum

$$A_n = \sum_{j=1}^n S(t_j).$$

Key point

- ▶ *For path-dependent products, the state variable is often larger than spot alone.*

Validate the Path, Not Only the Terminal Distribution

A robust implementation should identify the exact path feature entering the payoff and validate that feature directly.

- ▶ **Barrier products:** payoff depends on whether/when the barrier is crossed. Check missed crossings between time steps.
- ▶ **Asian products:** payoff depends on prices sampled along the path. Check averaging dates and weights.
- ▶ **Lookbacks:** payoff depends on the path maximum or minimum. Check whether discrete simulation misses extrema.
- ▶ **Autocallables and cliquets:** payoff depends on scheduled observations and past states. Check trigger dates, resets, and state updates.

Then perform convergence tests as

$$\Delta t \downarrow 0$$

and compare alternative path constructions where possible.

Practical principle

- ▶ *A simulation may reproduce terminal asset prices and vanilla option values while still generating inaccurate paths. Path-dependent derivatives therefore require separate validation of the simulated path between valuation date and maturity.*

Learn more

- ▶ For simulation and pricing of path-dependent derivatives, see [Lecture CF13: Lecture 13—Exotic Derivatives](#), and [Lecture CF12](#): for forward-start options.

Quant Rule #20

Before pricing a path-dependent derivative:

- ▶ write the payoff explicitly as a functional of the path;
- ▶ identify the required path statistics and observation dates;
- ▶ distinguish discrete from continuous monitoring;
- ▶ test time-grid and monitoring convergence;
- ▶ validate path features, not only terminal distributions.

Quant Rule #20

- ▶ If the payoff depends on the path, the simulation must preserve the path features that determine the payoff.

Matching only the terminal distribution is not enough.

correct $S(T)$ $\nRightarrow$ correct path-dependent price.

Final message

- ▶ *Never discard information that appears explicitly in the payoff.*

Quant Mistake #21: Treating a Forward Rate as a Forecast of the Future Rate

The incorrect belief

- ▶ If today's forward rate for the future period $[T_1, T_2]$ is 4%, then the market's expected future rate must also be 4%.

The mistake

- ▶ *A forward rate is a market-implied rate extracted from today's prices. It reflects market information and expectations, but also risk premia and other pricing effects. Therefore, it need not equal the real-world expected future rate.*

For a simple forward rate,

$$L(t; T_1, T_2) = \frac{1}{\alpha} \left(\frac{P(t, T_1)}{P(t, T_2)} - 1 \right), \text{ where } \alpha = \text{DCF}(T_1, T_2).$$

In general,

$$L(t; T_1, T_2) \neq \mathbb{E}^{\mathbb{P}} [L(T_1; T_1, T_2) | \mathcal{F}(t)].$$

The consequence

- ▶ Using the forward curve as a literal forecast can lead to incorrect economic scenarios, expected-rate paths, trading signals and risk assessments.

What Does the Forward Rate Actually Represent?

- ▶ Consider $t < T_1 < T_2$.
- ▶ The simple forward rate for the future accrual period $[T_1, T_2]$ is

$$L(t; T_1, T_2) = \frac{1}{\alpha} \left(\frac{P(t, T_1)}{P(t, T_2)} - 1 \right).$$

- ▶ This relation follows from today's discount factors.
- ▶ Equivalently,

$$P(t, T_1) = P(t, T_2) (1 + \alpha L(t; T_1, T_2)).$$

- ▶ $P(t, T_1)$ and $P(t, T_2)$ are market-implied prices observed today.
- ▶ The forward rate is therefore implied by today's relative bond prices.
- ▶ It is the rate consistent with no-arbitrage pricing of cash flows over the future period $[T_1, T_2]$.

Key point

- ▶ *The forward rate answers a pricing question:*

"Which future-period rate is consistent with today's market prices?"

That is different from asking:

"Which rate do we believe will actually be observed at T_1 ?"

The Future Fixing Is Still Random

- ▶ At time $t < T_1$, the future fixing

$$L(\textcolor{red}{T}_1; T_1, T_2)$$

is **not yet known**.

- ▶ The forward rate

$$L(\textcolor{red}{t}; T_1, T_2)$$

is **known today** because it is calculated from today's market prices.

- ▶ Thus we must distinguish:

$L(t; T_1, T_2)$	known at time t	market-implied forward
$L(T_1; T_1, T_2)$	unknown at time t	future realized fixing.

- ▶ At the fixing date T_1 , the uncertainty disappears and

$$L(T_1; T_1, T_2)$$

becomes observable.

- ▶ Before T_1 , however, many future fixing values are possible.

Important

- ▶ *Knowing today's forward rate is not the same as knowing the future fixing. A market-implied price today does not reveal one future realized state.*

Under Which Measure Is the Forward Rate an Expectation?

- ▶ The distinction becomes precise when the probability measure is specified.
- ▶ Recall

$$1 + \alpha L(t; T_1, T_2) = \frac{P(t, T_1)}{P(t, T_2)}.$$

- ▶ Use the bond $P(t, T_2)$ as numeraire.
- ▶ Under the corresponding T_2 -forward measure $\mathbb{Q}^{T_2}$, the ratio

$$\frac{P(t, T_1)}{P(t, T_2)} \text{ is a martingale.}$$

- ▶ Therefore,

$$L(t; T_1, T_2)$$

is also a martingale under $\mathbb{Q}^{T_2}$, and

$$L(t; T_1, T_2) = \mathbb{E}^{\mathbb{Q}^{T_2}} [L(T_1; T_1, T_2) | \mathcal{F}(t)].$$

Central distinction

- ▶ *The forward rate is an expectation under its natural pricing measure.
This does not make it the expected future rate under the real-world measure $\mathbb{P}$.*

$$\mathbb{E}^{\mathbb{Q}^{T_2}} [\text{future rate}] \neq \mathbb{E}^{\mathbb{P}} [\text{future rate}] \quad \text{in general.}$$

The Same Rate Has Different Drifts under Different Measures

- Under its natural forward measure, a simple rate may be modeled schematically as

$$dL(t) = \sigma_L(t)dW^{T_2}(t),$$

so its drift is zero.

- But under the real-world measure $\mathbb{P}$, the same process generally has dynamics of the form

$$dL(t) = \mu_L^{\mathbb{P}}(t)dt + \sigma_L(t)dW^{\mathbb{P}}(t).$$

- The real-world drift $\mu_L^{\mathbb{P}}(t)$ need not vanish.
- Therefore,

$$\mathbb{E}^{\mathbb{P}}[L(T_1) | \mathcal{F}(t)]$$

need not equal today's forward rate.

- Define the difference

$$\pi(t; T_1, T_2) := \mathbb{E}^{\mathbb{P}}[L(T_1; T_1, T_2) | \mathcal{F}(t)] - L(t; T_1, T_2).$$

- The quantity π represents the wedge between the physical forecast and the market-implied forward rate.

Important

- There is no universal reason for this wedge to be zero or to have one fixed sign.*

Example: Forward Rate versus Real-World Forecast

- Suppose today's discount factors are

$$P(0, 1) = 0.9600, \quad P(0, 2) = 0.9150, \quad \text{and let } \alpha = 1.$$

- The one-year forward rate for the period $[1, 2]$ is

$$\begin{aligned} L(0; 1, 2) &= \frac{P(0, 1)}{P(0, 2)} - 1 \\ &= \frac{0.9600}{0.9150} - 1 \\ &\approx \boxed{4.92\%}. \end{aligned}$$

- Now suppose an economic forecasting model estimates

$$\mathbb{E}^{\mathbb{P}} [L(1; 1, 2)] = 4.20\%.$$

- Then

$$\pi = 4.20\% - 4.92\% = -0.72\%.$$

Is there a contradiction?

- No.

The 4.92% forward rate is implied by today's traded bond prices.

The 4.20% number is a real-world statistical forecast.

They answer different questions and may therefore differ substantially.

A Correct Pricing Model Can Still Give a Poor Forecast

- ▶ Suppose a model is perfectly consistent with today's market curve and gives

$$L(t; T_1, T_2) = 4.92\%.$$

- ▶ This does not imply that the model predicts

$$L(T_1; T_1, T_2) = 4.92\%$$

as the most likely or average real-world outcome.

- ▶ The pricing model may be constructed under

$$\mathbb{Q} \quad \text{or} \quad \mathbb{Q}^{T_2},$$

while forecasting requires a specification under

$$\mathbb{P}.$$

- ▶ Thus it is possible to have simultaneously:

correct arbitrage-free pricing and
incorrect interpretation as a real-world forecast.

Important distinction

- ▶ *The mistake does not require an incorrect pricing formula.
The pricing calculation may be completely correct; the error occurs when its output is given the wrong economic interpretation.*

Where the Mistake Matters in Practice

Treating the forward curve as an expected future yield curve can affect:

- ▶ **economic forecasting:** interpreting market forwards as consensus predictions;
- ▶ **asset–liability management:** using forwards as expected refinancing or reinvestment rates;
- ▶ **scenario generation:** constructing a “base-case” future rate path directly from today’s forward curve;
- ▶ **trading strategies:** calling a future rate “cheap” or “expensive” only because it differs from today’s forward;
- ▶ **risk analysis:** confusing a pricing-measure term structure with a physical distribution of future rates.

Practical warning

- ▶ *A forward curve is an essential input for arbitrage-free pricing.
A forward curve provides a market-implied view of future rates, but converting it into a real-world expected path requires assumptions about risk premia and physical dynamics.*

When Can a Forward Rate Be Used as a Forecast?

- ▶ Using the forward rate as an approximation to a future expected rate may be reasonable in a particular application.
- ▶ But this is an **additional modeling assumption**, not a consequence of the forward-rate definition.
- ▶ For example, one may assume approximately

$$\mathbb{E}^{\mathbb{P}} [L(T_1; T_1, T_2) | \mathcal{F}(t)] \approx L(t; T_1, T_2)$$

when the estimated physical expectation is close to the market-implied forward rate.

- ▶ A robust analysis should distinguish:

market-implied forward curve $\neq$ real-world expected future curve $\neq$ future realized curve.

Key lesson

- ▶ *The equality may be assumed as an approximation.*

Learn more

- ▶ For forward rates, FRAs and floating-rate products, see [Lecture FE5: Interest Rate Products](#).
- ▶ For forward measures and measure-dependent rate dynamics, see [Lecture FE11: Market Model and Convexity Adjustments](#).

Quant Rule #21

Whenever a forward rate is interpreted economically, ask:

- ▶ Is this quantity implied by current market prices?
- ▶ Under which probability measure is an expectation being taken?
- ▶ Are we performing arbitrage-free pricing or real-world forecasting?
- ▶ What assumptions are being made about risk premia?
- ▶ Are we distinguishing today's forward from the future realized fixing?

Quant Rule #21

- ▶ A forward rate is a market-implied pricing quantity.
It equals the expected future fixing under its appropriate forward measure, but it is not, in general, the expected future fixing under the real-world measure.

forward rate $\neq$ real-world forecast $\neq$ future realization.
--

Final message

- ▶ *Never call a forward rate the market's forecast of a future rate without specifying the probability measure and the assumptions that justify that interpretation.*

Quant Mistake #22: Treating a Par Swap Rate as a Zero Rate

The incorrect belief

- ▶ If the market quotes a five-year par swap rate of 5%, then the five-year zero rate is also 5%.

The mistake

- ▶ *A par swap rate and a zero rate describe different financial quantities. A zero rate determines one discount factor at one maturity. A par swap rate depends on discount factors across all swap payment dates. With a non-flat yield curve, these rates will generally differ.*

par swap rate $\neq$ zero rate

in general; they may coincide in special cases, such as a flat curve under matching conventions.

The consequence

- ▶ Treating swap quotes as zero rates can produce incorrect discount factors, incorrect forward rates and a curve that does not reproduce the market swap quotes from which it was constructed.

A Zero Rate Describes One Discount Factor

- ▶ To isolate the issue, assume throughout this example that:
 - ▶ all swap payment dates coincide exactly with curve pillars;
 - ▶ all zero rates use the same annual-compounding convention;
 - ▶ no interpolation is required.
- ▶ For an annually compounded zero rate z_n at maturity $T_n = n$,

$$P(0, T_n) = \frac{1}{(1 + z_n)^n}.$$

- ▶ Thus z_n describes one terminal discount factor:

$$z_n \longleftrightarrow P(0, T_n).$$

- ▶ For example,

$$z_5 \longleftrightarrow P(0, T_5).$$

Key point

- ▶ *A five-year zero rate contains information about the discount factor at year five. It does not directly describe the value of cash flows paid at years one, two, three and four.*

A Par Swap Rate Depends on the Entire Fixed Leg

- ▶ Consider a **spot-starting** swap with payment dates $T_1 < T_2 < \dots < T_n$, and accruals α_i .
- ▶ The present value of the fixed leg with fixed rate S_n is

$$PV_{\text{fixed}} = N S_n \sum_{i=1}^n \alpha_i P(0, T_i).$$

- ▶ In a single-curve setting, the floating leg is worth

$$PV_{\text{float}} = N \sum_{i=1}^n \alpha_i F_i P(0, T_i) = N(1 - P(0, T_n)), \quad F_i = \frac{1}{\alpha_i} \left(\frac{P(0, T_{i-1})}{P(0, T_i)} - 1 \right).$$

- ▶ At the par swap rate,

$$PV_{\text{fixed}} = PV_{\text{float}}.$$

- ▶ Therefore,

$$S_n = \frac{1 - P(0, T_n)}{\sum_{i=1}^n \alpha_i P(0, T_i)}.$$

Central distinction

- ▶ The zero rate z_n determines one discount factor $P(0, T_n)$.
The par rate S_n depends on **all discount factors** appearing in the fixed-leg annuity.

Example: A 5% Par Rate Is Not a 5% Zero Rate

- ▶ Consider a five-year annual-pay swap.
- ▶ Suppose the first four annually compounded zero rates are already known:

i	z_i	$P(0, T_i)$
1	2.00%	0.98039
2	2.50%	0.95181
3	3.00%	0.91514
4	3.50%	0.87144

- ▶ Suppose the market quotes the five-year par swap rate

$$S_5 = 5.00\%.$$

- ▶ Since $\alpha_i = 1$, the par condition is

$$0.05(P_1 + P_2 + P_3 + P_4 + P_5) = 1 - P_5.$$

- ▶ Solving for the unknown five-year discount factor gives

$$P_5 = \frac{1 - 0.05(P_1 + P_2 + P_3 + P_4)}{1.05} \approx 0.77530.$$

Important

- ▶ *The swap quote gives an equation for P_5 .
It does not say that the five-year zero rate equals 5%.*

What Is the Five-Year Zero Rate?

- From the previous slide,

$$P_5 = P(0, T_5) \approx 0.77530.$$

- Using the same annual-compounding convention,

$$P_5 = \frac{1}{(1 + z_5)^5}.$$

- Therefore,

$$\begin{aligned} z_5 &= P_5^{-1/5} - 1 \\ &= (0.77530)^{-1/5} - 1 \\ &\approx \boxed{5.222\%}. \end{aligned}$$

- But the market par swap rate is

$$\boxed{S_5 = 5.000\%}.$$

- Hence,

$$\boxed{S_5 = 5.000\% \quad \neq \quad z_5 = 5.222\%}.$$

Why are they different?

- The five-year zero rate describes only $P(0, T_5)$.*

The five-year par rate also depends on the discount factors at T_1, T_2, T_3, T_4 .

No interpolation and no change of compounding convention is involved in this difference.

Treating the Par Rate as the Zero Rate Breaks the Swap Price

- Suppose we incorrectly set

$$z_5^{\text{wrong}} = S_5 = 5\%.$$

- Then the five-year discount factor becomes

$$P_5^{\text{wrong}} = \frac{1}{1.05^5} \approx 0.78353.$$

- Using this discount factor together with the correct $P_1, \dots, P_4$, the par rate implied by the resulting curve is

$$S_5^{\text{curve}} = \frac{1 - P_5^{\text{wrong}}}{P_1 + P_2 + P_3 + P_4 + P_5^{\text{wrong}}} \\ \approx 4.808\%.$$

- But the market quote was

$$5.000\%.$$

- Thus,

$$z_5^{\text{wrong}} = S_5 \implies S_5^{\text{curve}} \neq S_5^{\text{market}}.$$

Immediate diagnostic

- If a swap quote is used to determine the curve, the resulting curve must reproduce that swap quote.*
Simply copying the par rate into the zero curve fails this test.

The Error Can Be Material

- ▶ Consider the same five-year swap with notional

$$N = 100 \text{ million}$$

and fixed rate

$$K = 5\%.$$

- ▶ With the correctly constructed curve, K is the market par rate and

$$V_{\text{correct}} = 0.$$

- ▶ Using instead

$$P_5^{\text{wrong}} = 0.78353,$$

the value of a payer swap is

$$V_{\text{wrong}} = N \left[1 - P_5^{\text{wrong}} - K \sum_{i=1}^5 P_i^{\text{wrong}} \right]$$

$$\approx -0.864 \text{ million.}$$

- ▶ Therefore, the absolute pricing error is approximately

864,000.

Lesson

- ▶ *The mistake is not merely a difference in how rates are quoted. Confusing a par rate with a zero rate changes discount factors, forward rates and the value of the instruments priced from the curve.*

Quant Rule #22

Whenever a market swap rate is used in yield-curve construction:

- ▶ distinguish the par swap rate from the zero rate;
- ▶ write the full swap pricing equation;
- ▶ use all relevant fixed-leg discount factors;
- ▶ solve for the unknown curve quantity;
- ▶ verify that the resulting curve reproduces the market swap rate;
- ▶ keep the compounding convention explicit.

Quant Rule #22

- ▶ A par swap rate is the coupon rate that makes a multi-cash-flow swap worth zero. It is not the zero rate at the swap maturity.

par rate = multi-cash-flow quantity, zero rate = single-maturity quantity.

Final message

- ▶ *Never copy a market par swap rate directly into a zero-rate curve. Use the swap quote through the pricing equation that defines it.*

Learn more

- ▶ For yield-curve construction, discount factors, forward rates and swap rates, see [Lecture FE6: Yield Curve Construction](#).

Quant Mistake #23: Using One Yield Curve for Both Forecasting and Discounting

The incorrect belief

- ▶ Build one yield curve and use it both to forecast future floating rates and to discount every cash flow.

The mistake

- ▶ *Forecasting and discounting are different economic tasks.
The curve used to project a floating index need not be the curve that determines the present value of the resulting cash flow.*

For a future floating coupon, distinguish:

- ▶ the projection curve, linked to the contractual index;
- ▶ the discount curve, linked to collateral and funding conventions.

curve for expected cash flow $\neq$ curve for present value

in general.

Projection and Discounting Answer Different Questions

- ▶ Consider a floating coupon over

$$[T_{i-1}, T_i]$$

paid at T_i .

- ▶ Before fixing, its projected cash flow is approximately

$$N\alpha_i \left(F_i^{proj}(t) + s \right),$$

where $F_i^{proj}(t)$ is obtained from the curve associated with the contractual floating index.

- ▶ Its present value is then

$$PV_i(t) = P^{disc}(t, T_i) N\alpha_i \left(F_i^{proj}(t, T_{i-1}, T_i) + s \right).$$

- ▶ Projection determines the future coupon amount.
- ▶ Discounting determines today's value of that amount.

Key distinction

- ▶ *Using the same curve for both operations is a modeling assumption, not an identity.*

Why the Single-Curve Cancellation Breaks

- ▶ In a classical single-curve framework,

$$F_i(t) = \frac{1}{\alpha_i} \left(\frac{P(t, T_{i-1})}{P(t, T_i)} - 1 \right)$$

- ▶ Using the same curve for discounting, significantly simplifies **float leg** valuation:

$$\sum_{i=1}^N P(t, T_i) \alpha_i F_i(t) = \sum_{i=1}^N P(t, T_i) \left(\frac{P(t, T_{i-1})}{P(t, T_i)} - 1 \right) = P(t, T_0) - P(t, T_N).$$

- ▶ Multi-curve world: projection $\neq$ discounting

$$F_i^{proj}(t) = \frac{1}{\alpha_i} \left(\frac{P^{proj}(t, T_{i-1})}{P^{proj}(t, T_i)} - 1 \right),$$

but valuation uses $P^{disc}(t, T_i)$.

- ▶ Hence

$$P^{disc}(t, T_i) \alpha_i F_i^{proj}(t) = P^{disc}(t, T_i) \left(\frac{P^{proj}(t, T_{i-1})}{P^{proj}(t, T_i)} - 1 \right),$$

and there is **nothing to cancel** because $P^{disc}(t, T_i) \neq P^{proj}(t, T_i)$.

Key takeaway

- ▶ *The telescoping identity relies on using the same curve for both forward-rate construction and discounting. Once projection and discount curves differ, the floating leg no longer collapses to $P(t, T_0) - P(t, T_N)$.*

Concrete Example: The Wrong Curve Changes the Coupon Value

- ▶ Consider one future floating coupon with

$$N = 100 \text{ million}, \quad \alpha = 0.25.$$

- ▶ Suppose the contractual index projection curve gives

$$F^{proj} = 4.50\%,$$

while the discount curve implies a same-period forward of only

$$F^{disc} = 3.80\%.$$

- ▶ Let the correct discount factor be

$$P^{disc}(t, T) = 0.98.$$

- ▶ Using the proper projection curve,

$$PV_{correct} = 0.98 \times 100\text{m} \times 0.25 \times 0.045 \approx 1.1025\text{m}.$$

- ▶ Using the discount curve to forecast as well gives

$$PV_{wrong} \approx 0.9310\text{m}.$$

- ▶ The difference is about

171,500.

The Error Propagates into Swaps and Risk

For a floating leg,

$$PV_{float} = \sum_{i=1}^n P^{disc}(t, T_i) N_i \alpha_i \left(F_i^{proj}(t) + s \right).$$

Using one inappropriate curve can distort:

- ▶ par swap rates and basis spreads;
- ▶ forward rates and FRA values;
- ▶ curve sensitivities and hedge ratios;
- ▶ calibration of caps, swaptions and market models.

The risk decomposition also changes:

$$PV \text{ sensitivity} = \text{projection-curve risk} + \text{discount-curve risk} + \dots$$

Practical warning

- ▶ *A price can look reasonable while the risk is assigned to the wrong curve, leading to poor hedges even when the valuation error appears small.*

Quant Rule #23

Before valuing an interest-rate product:

- ▶ identify the contractual floating index;
- ▶ choose the corresponding projection curve;
- ▶ identify the collateral/funding convention;
- ▶ choose the corresponding discount curve;
- ▶ compute and report risk by curve rather than hiding it in one bucket.

Quant Rule #23

- ▶ Forecasting and discounting are separate operations.

Use the curve associated with the floating index to project cash flows and the curve implied by the collateral/funding framework to discount them.

A single-curve framework is a special case that should be justified by the market structure, not assumed by default.

one instrument $\nrightarrow$ one yield curve.

Final message

- ▶ *Do not force a single curve onto a market structure that prices projection and discounting differently.*

Quant Mistake #24: Choosing Yield-Curve Interpolation Only Because It Looks Smooth

The incorrect belief

- ▶ Choose the interpolation method that produces the smoothest-looking yield curve between market pillars.

The mistake

- ▶ *Visual smoothness is not the objective of yield-curve construction. An interpolation can look excellent in zero-rate space while producing unstable discount factors, artificial forward-rate oscillations or poor risk behaviour.*

A good interpolation should be judged by:

- ▶ exact or controlled repricing of calibration instruments;
- ▶ financially sensible discount and forward curves;
- ▶ stability under market bumps;
- ▶ appropriate locality and extrapolation behaviour.

smooth picture $\neq$ good curve construction.

The Same Market Nodes Can Produce Different Curves

- ▶ Suppose the bootstrap determines curve values at market pillars

$$T_1, T_2, \dots, T_n.$$

- ▶ Between those dates, **interpolation is needed**.
- ▶ But one may interpolate different quantities:
 - ▶ zero rates $z(T)$;
 - ▶ discount factors $P(0, T)$;
 - ▶ log-discount factors $\log P(0, T)$;
 - ▶ instantaneous or discrete forward rates.
- ▶ These choices are not equivalent because interpolation acts before the transformations

$$P(0, T) = e^{-z(T)T}, \text{ and } F(0; T_1, T_2) = \frac{1}{\alpha} \left(\frac{P(0, T_1)}{P(0, T_2)} - 1 \right),$$

and, under continuous compounding,

$$f(0, T) = -\frac{\partial}{\partial T} \log P(0, T) = z(T) + Tz'(T).$$

Key point

- ▶ *A curve that is smooth in one representation need not be well behaved in another representation.*

Smooth Zero Rates Can Hide Wild Forward Rates

- ▶ The instantaneous forward rate depends on the derivative of the zero curve:

$$f(0, T) = z(T) + Tz'(T).$$

- ▶ Therefore, small interpolation wiggles in $z(T)$ can be amplified through

$$Tz'(T).$$

- ▶ high-order spline may pass smoothly through every calibration node while overshooting between them.
- ▶ Consequences can include:
 - ▶ artificial humps and dips in forward rates;
 - ▶ unstable short-period forward projections;
 - ▶ implausible local curve shapes between liquid pillars;
 - ▶ exaggerated sensitivities to individual market quotes.

Central lesson

- ▶ *Inspect the economically relevant derived quantities- especially forwards and discount factors- not only the interpolated object itself.*

Concrete Example: A Smooth Spline Can Create Artificial Forwards

- ▶ Consider illustrative continuously compounded zero-rate nodes:

T	0.5	1	2	3	5	7	10
$z(T)$	2.0%	2.5%	4.0%	2.0%	3.0%	2.5%	3.0%

- ▶ A natural **cubic spline** through these points is perfectly smooth in $z(T)$.
- ▶ But the implied instantaneous forward

$$f(0, T) = z(T) + Tz'(T)$$

can reach approximately

−4.3%

near $T \approx 2.8$,

and approximately

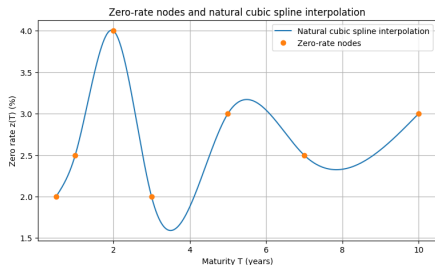
8.2%

near $T \approx 4.6$.

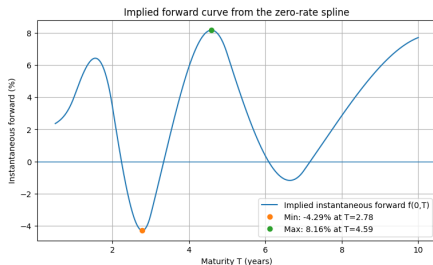
Lesson

- ▶ *All zero-rate nodes lie between 2% and 4%, yet the interpolated forward curve develops much larger oscillations between the nodes.*
The spline is smooth- but the financial object it implies may be poor.

Smooth Zero Rates Can Produce Artificial Forward Rates



Natural cubic spline interpolation of the zero-rate nodes.



Implied instantaneous forward curve

$$f(0, T) = z(T) + Tz'(T).$$

Key takeaway

- ▶ Although the zero-rate curve looks smooth and all observed zero rates lie between roughly 2% and 4%, the implied forward curve reaches about

−4.3% and 8.2%.

Smooth interpolation of zero rates does not guarantee economically reasonable forward rates.

Interpolation Changes Prices and Risk Between Pillars

- ▶ For a cash flow C paid at a **non-pillar** date T , its value is

$$PV = C \cdot P(0, T).$$

- ▶ Thus interpolation directly affects every off-node discount factor.
- ▶ For floating products, it also affects forwards such as

$$F(0; T_1, T_2) = \frac{1}{\alpha} \left(\frac{P(0, T_1)}{P(0, T_2)} - 1 \right).$$

- ▶ The choice can therefore change:
 - ▶ swap, FRA and basis valuations;
 - ▶ off-cycle cash flows and stub periods;
 - ▶ bucketed DV01 and hedge ratios;
 - ▶ calibration stability under small quote bumps.

Risk warning

- ▶ *An interpolation method can create nonlocal risk: bumping one market pillar may move a large region of the curve because of the interpolation itself.*

Choose Interpolation by Financial Criteria

- ▶ There is no universally best interpolation method.
- ▶ Common choices include:
 - ▶ linear interpolation of zero rates;
 - ▶ log-linear interpolation of discount factors;
 - ▶ spline-based methods;
 - ▶ monotone-convex or other forward-aware constructions.
- ▶ A useful validation checklist is:
 - ▶ Reprice the calibration instruments.
 - ▶ Inspect discount factors and forward rates between pillars.
 - ▶ Bump each market quote and inspect curve/risk locality.
 - ▶ Test short-end, long-end and extrapolation behaviour.
 - ▶ Compare economically relevant prices under alternative methods.

Correct principle

- ▶ *Choose the interpolation whose mathematical properties match the intended pricing and risk-management use- not the one that merely looks prettiest.*

Learn more

- ▶ For yield-curve construction and interpolation choices, see [Lecture FE6: Construction of Yield Curve and Multi-Curve](#).

Quant Rule #24

Before approving a curve interpolation, test:

- ▶ calibration-instrument repricing;
- ▶ discount-factor and forward-rate behaviour;
- ▶ sensitivity locality under pillar bumps;
- ▶ robustness of off-node prices and hedge ratios;
- ▶ extrapolation outside the liquid market region.

Quant Rule #24

- ▶ Yield-curve interpolation is part of the pricing model.
Judge it by financial consistency, stability and risk behaviour- not by visual smoothness alone.

smooth interpolation $\nrightarrow$ sound forwards or stable risk.

Final message

- ▶ *Always inspect what the interpolation implies between the market pillars, not only how beautifully it passes through them.*

Quant Mistake #25: Using a Lognormal Swap-Rate Model in a Negative-Rate Environment

The incorrect belief

- ▶ Swap rates are usually modeled as lognormal, so the same model can be used regardless of the level of rates.

The mistake

- ▶ *A pure lognormal model imposes a sign restriction on the modeled rate. Starting from a positive swap rate, it cannot cross zero and become negative.*

This becomes problematic when:

- ▶ observed swap rates are near or below zero;
- ▶ option strikes include zero or negative values;
- ▶ scenarios require realistic probability of crossing zero.

model support must contain the market states you need to represent.

Why Lognormal Dynamics Cannot Cross Zero

- ▶ Under the swap annuity measure, a simplified lognormal swap-rate model is

$$dS(t) = \sigma S(t) dW^A(t).$$

- ▶ Its solution is

$$S(T) = S(0) \exp \left(-\frac{1}{2} \sigma^2 T + \sigma W^A(T) \right).$$

- ▶ Therefore, if

$$S(0) > 0,$$

then

$$S(T) > 0 \quad \text{almost surely.}$$

Structural restriction

- ▶ *The model assigns zero probability to negative future swap rates. This is not a calibration issue- it is built into the state space of the model.*

Concrete Example: The Missing Probability Mass

- Suppose the one-year forward swap rate is

$$S(0) = 0.25\%.$$

- A normal model with annual normal volatility

$$\sigma_N = 1.00\%$$

gives

$$S(T) = S(0) + \sigma_N W^A(T).$$

- For $T = 1$,

$$\begin{aligned} \mathbb{Q}[S(T) < 0] &= \Phi\left(-\frac{0.25\%}{1.00\%}\right) \\ &\approx 40.1\%. \end{aligned}$$

- A positive lognormal model gives instead

$$\mathbb{Q}[S(T) < 0] = 0.$$

Lesson

- The disagreement is not numerical noise. The two models allow fundamentally different future states.*

The Black Formula Can Become the Wrong Tool

- ▶ A payer swaption payoff at expiry T is proportional to

$$(S(T) - K)^+.$$

- ▶ The classical Black swaption formula is based on lognormal swap-rate dynamics and quantities such as

$$\log\left(\frac{F}{K}\right).$$

- ▶ If the market contains

$$F \leq 0 \quad \text{or} \quad K \leq 0,$$

the standard lognormal Black setup is no longer naturally compatible with the market state.

- ▶ For example,

$$F = -0.50\%, \quad K = 0.00\%$$

cannot be handled by simply inserting the rates into the ordinary Black formula.

Common mistake

- ▶ *Do not force a lognormal formula onto negative inputs by ad hoc clipping or by silently replacing the rate with a small positive number.*

Two Common Alternatives

Normal / Bachelier model

- Model the swap rate additively:

$$dS(t) = \sigma_N dW^A(t).$$

Then $S(t)$ can move across zero naturally.

Shifted-lognormal model

- Choose a displacement λ and define

$$X(t) := S(t) + \lambda > 0, \quad dX(t) = \sigma_L X(t) dW^A(t).$$

Hence the modeled swap rate satisfies

$$S(t) > -\lambda.$$

Example:

$$S(0) = -0.50\%, \quad \lambda = 2.00\%, \quad X(0) = 1.50\% > 0.$$

Important

- *The shift is part of the model: it changes the dynamics, **smile behavior and lower bound of the rate**.*

Choose the Model and Volatility Convention Consistently

- ▶ Normal, lognormal and shifted-lognormal models use different volatility coordinates.
 - ▶ Normal volatility is an absolute rate volatility.
 - ▶ Lognormal volatility scales moves relative to the rate level.
 - ▶ Shifted-lognormal volatility scales moves relative to $S + \lambda$.
- ▶ Therefore, the same numerical volatility value **does not** represent the same risk across these models.
- ▶ A robust implementation should:
 - ▶ use the market quoting convention explicitly;
 - ▶ convert quotes consistently when comparing models;
 - ▶ test prices and Greeks near zero and for negative strikes/rates;
 - ▶ stress whether the model's state-space boundary is economically sensible.

Learn more

- ▶ For swap-rate models and swaption pricing, see [Lecture FE7: Pricing of Swaptions and Negative Interest Rates](#).

Quant Rule #25

Before choosing a swap-rate model, ask:

- ▶ Can the model generate the rate levels observed in the market?
- ▶ Can it cross zero when the product or scenario requires it?
- ▶ Are the strike domain and volatility quoting convention compatible?
- ▶ Is a normal or shifted-lognormal specification more appropriate?
- ▶ Are prices and Greeks stable near the model boundary?

Quant Rule #25

- ▶ Never use a lognormal rate model merely because it is standard. The support of the model must be compatible with the support required by the market and the product.

A model that cannot reach the market state cannot price it reliably.

Final message

- ▶ *Choose the distributional assumption first from the economics of the rate environment- then choose the pricing formula that matches it.*

Quant Mistake #26: Assuming that All Forward Rates Are Driftless under One Measure

The incorrect belief

- ▶ In a Forward (Libor) Market Model, every forward rate is a martingale, so all rates can be simulated with zero drift under one common pricing measure.

The mistake

- ▶ *Each forward rate is naturally driftless under its own forward measure. Changing to a common measure changes the Brownian motion and creates drift terms.*

For the forward rate over $[T_i, T_{i+1}]$,

$$L_i(t) := L(t, T_i, T_{i+1}) = \frac{1}{\alpha_i} \left(\frac{P(t, T_i)}{P(t, T_{i+1})} - 1 \right).$$

driftless under its own measure $\neq$ driftless under every measure.

Every Forward Rate Has a Natural Measure

- ▶ Use the bond maturing at T_{i+1} as numeraire.
- ▶ Under the corresponding forward measure

$$\mathbb{Q}^{T_{i+1}},$$

the ratio

$$\frac{P(t, T_i)}{P(t, T_{i+1})}$$

is a martingale.

- ▶ Therefore $L_i(t) := L(t, T_i, T_{i+1})$ is a martingale under $\mathbb{Q}^{T_{i+1}}$, and a lognormal specification may be written as

$$\frac{dL_i(t)}{L_i(t)} = \sigma_i(t) \cdot dW^{T_{i+1}}(t).$$

- ▶ But L_{i+1} is naturally driftless under

$$\mathbb{Q}^{T_{i+2}},$$

not under $\mathbb{Q}^{T_{i+1}}$ in general.

Key point

- ▶ *Different forward rates generally have different natural numeraires and therefore different natural martingale measures.*

Under One Common Measure, Drifts Reappear

- ▶ Suppose all rates are simulated under the terminal measure

$$\mathbb{Q}^{T_{N+1}}.$$

- ▶ For a standard lognormal Forward Market Model,

$$\frac{dL_i(t)}{L_i(t)} = - \sum_{j=i+1}^N \frac{\alpha_j L_j(t)}{1 + \alpha_j L_j(t)} \sigma_i(t) \cdot \sigma_j(t) dt + \sigma_i(t) \cdot d\mathbf{W}^{T_{N+1}}(t).$$

- ▶ The drift depends on:
 - ▶ later forward rates L_j ;
 - ▶ accrual factors α_j ;
 - ▶ volatilities and correlations through $\sigma_i \cdot \sigma_j$.

Interpretation

- ▶ *The drift is the compensation required by the change of numeraire.*
- ▶ *Setting it to zero under the terminal measure changes the model.*
- ▶ *Zero drift may be also interpreted as “assumption” about **zero-correlation** between forward rates.*

Concrete Example: The Missing Drift Is Not Zero

- Consider two consecutive forward rates:

$$L_1 = L(t; T_1, T_2), \quad L_2 = L(t; T_2, T_3),$$

and simulate both under the terminal measure $\mathbb{Q}^{T_3}$.

- Under $\mathbb{Q}^{T_3}$, L_2 is driftless, but L_1 is not.
- Take

$$\begin{aligned} L_1 &= 5\%, & L_2 &= 5\%, & \alpha_2 &= 0.5, \\ \sigma_1 &= 40\%, & \sigma_2 &= 50\%, & \rho_{12} &= 0.9. \end{aligned}$$

- The relative drift of L_1 is

$$\mu_1 = -\frac{\alpha_2 L_2}{1 + \alpha_2 L_2} \sigma_1 \sigma_2 \rho_{12} \approx -0.00439.$$

- Hence, initially,

$$\frac{dL_1}{L_1} = \underbrace{-0.439\% dt}_{\text{drift}} + \underbrace{40\% dW^{T_3}}_{\text{diffusion}}.$$

For $L_1 = 5\%$, the drift in the rate level is approximately

$$5\% \times (-0.439\%) \approx -2.2 \text{ bp per year.}$$

Lesson

- Changing from L_1 's natural measure $\mathbb{Q}^{T_2}$ to the common terminal measure $\mathbb{Q}^{T_3}$ introduces a nonzero drift.

Why Setting Every Drift to Zero Is Inconsistent

- ▶ Suppose one writes under a single measure $\mathbb{Q}^*$

$$\frac{dL_i}{L_i} = \sigma_i \cdot dW^* \quad \text{for every } i.$$

- ▶ Then all forward rates are being forced to be martingales under the same numeraire.
- ▶ But the bond ratios defining different rates are associated with different numeraires:

$$1 + \alpha_i L_i(t) = \frac{P(t, T_i)}{P(t, T_{i+1})}.$$

- ▶ A change of numeraire changes the drift unless special covariance terms vanish.
- ▶ Consequences of ignoring this can include:
 - ▶ inconsistent bond-price dynamics;
 - ▶ biased caplet, swaption or exotic prices;
 - ▶ incorrect Monte Carlo evolution of the forward curve;
 - ▶ wrong convexity effects and hedge ratios.

Central lesson

- ▶ *Martingale is always a statement relative to a chosen numeraire and measure.*

Choose the Measure First, Then Derive Every Drift

- ▶ A consistent market-model implementation proceeds in this order:
 - ▶ Choose a common simulation measure, for example a **terminal or spot measure**.
 - ▶ Specify the volatility structure

$$\sigma_i(t).$$

- ▶ Derive each rate drift from the chosen **numeraire and measure**.
- ▶ Simulate all rates consistently under that one measure.
- ▶ Validate martingale identities for quantities that should be martingales under the chosen numeraire.

Implementation warning

- ▶ *Do not copy the zero-drift SDE from each rate's own forward measure and use all of those SDEs simultaneously under one common measure.*

Learn more

- ▶ For forward measures, the Forward Market Model and measure changes, see [Lecture FE11: Market Model and Convexity Adjustments](#).

Quant Rule #26

Before simulating a family of forward rates:

- ▶ identify the numeraire and probability measure;
- ▶ know which rate is naturally driftless under which forward measure;
- ▶ derive all drifts after converting to the common simulation measure;
- ▶ include volatility and correlation terms consistently;
- ▶ test the relevant martingale and bond-reconstruction identities.

Quant Rule #26

- ▶ A forward rate is driftless under its own forward measure.
Under a different common measure, its drift generally does not vanish and must be obtained by the correct change of numeraire.

no measure $\implies$ no meaningful statement about drift.
--

Final message

- ▶ *Never say that a rate is “driftless” without also saying under which measure.*

Quant Mistake #27: Ignoring the Convexity Effect of a Payment Delay

The incorrect belief

- ▶ If a floating rate is fixed for the period $[T_{\text{st}}, T_{\text{end}}]$ but paid later at $T_{\text{pay}} > T_{\text{end}}$, just use the ordinary forward rate and discount the cash flow to T_{pay} .

The mistake

- ▶ *The natural measure for the forward rate is linked to T_{end} , while the delayed cash flow is naturally valued using the payment date T_{pay} .
Changing the numeraire generally changes the expected future rate.*

Therefore,

$$\mathbb{E}^{\mathbb{Q}^{T_{\text{pay}}}} [L(F; T_{\text{st}}, T_{\text{end}})] \neq L(t; T_{\text{st}}, T_{\text{end}})$$

in general.

payment delay $\implies$ possible convexity adjustment.

Without a Payment Delay, the Structure Is Simple

- ▶ Consider a floating rate fixed at T for the accrual period

$$[T_{\text{st}}, T_{\text{end}}]$$

with accrual factor α .

- ▶ If the coupon is paid at the rate maturity,

$$T_{\text{pay}} = T_{\text{end}},$$

then its value before fixing is

$$V(t) = N\alpha P(t, T_{\text{end}}) \mathbb{E}^{\mathbb{Q}^{T_{\text{end}}}} [L(T; T_{\text{st}}, T_{\text{end}}) | \mathcal{F}(t)].$$

- ▶ Under the T_{end} -forward measure,

$$\boxed{\mathbb{E}^{\mathbb{Q}^{T_{\text{end}}}} [L(T; T_{\text{st}}, T_{\text{end}}) | \mathcal{F}(t)] = L(t; T_{\text{st}}, T_{\text{end}}).}$$

- ▶ Hence

$$V(t) = N\alpha P(t, T_{\text{end}}) L(t; T_{\text{st}}, T_{\text{end}}).$$

Key point

- ▶ *The forward rate is a martingale under the measure associated with its natural maturity numeraire.*

A Delayed Payment Changes the Relevant Measure

- ▶ Now suppose the same rate is paid at

$$T_{\text{pay}} > T_{\text{end}}.$$

- ▶ The payoff is still

$$N\alpha L(T; T_{\text{st}}, T_{\text{end}}),$$

but it is **received at** T_{pay} .

- ▶ Therefore,

$$V(t) = N\alpha P(t, T_{\text{pay}}) \mathbb{E}^{\mathbb{Q}^{T_{\text{pay}}}} [L(T; T_{\text{st}}, T_{\text{end}}) \mid \mathcal{F}(t)].$$

- ▶ Define the convexity adjustment by

$$CA(t) := \mathbb{E}^{\mathbb{Q}^{T_{\text{pay}}}} [L(T; T_{\text{st}}, T_{\text{end}})] - L(t; T_{\text{st}}, T_{\text{end}}).$$

- ▶ Then

$$V(t) = N\alpha P(t, T_{\text{pay}}) [L(t; T_{\text{st}}, T_{\text{end}}) + CA(t)].$$

The mistake

- ▶ Using only $P(t, T_{\text{pay}})L(t; T_{\text{st}}, T_{\text{end}})$ implicitly assumes $CA(t) = 0$ without justification.

Where the Convexity Adjustment Comes From

- ▶ Let

$$Z_T = \frac{P(T, T_{\text{pay}})/P(T, T_{\text{end}})}{P(t, T_{\text{pay}})/P(t, T_{\text{end}})}$$

be the change-of-numeraire density from the T_{end} -forward measure to the T_{pay} -forward measure over $[t, F]$.

- ▶ Then

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}^{T_{\text{pay}}}} [L(T; T_{\text{st}}, T_{\text{end}})] &= \mathbb{E}^{\mathbb{Q}^{T_{\text{end}}}} [L(T; T_{\text{st}}, T_{\text{end}}) Z_T] \\ &= L(t; T_{\text{st}}, T_{\text{end}}) + \text{Cov}^{\mathbb{Q}^{T_{\text{end}}}} (L(T; T_{\text{st}}, T_{\text{end}}), Z_T). \end{aligned}$$

- ▶ Therefore,

$$\boxed{\text{CA}(t) = \text{Cov}^{\mathbb{Q}^{T_{\text{end}}}} (L(T; T_{\text{st}}, T_{\text{end}}), Z_F).}$$

- ▶ The adjustment depends on rate **volatility and correlation**.
- ▶ It increases with the economic relevance of the payment mismatch.
- ▶ **Its sign is model- and structure-dependent; it is not universally positive.**

Central lesson

- ▶ *The adjustment is a measure-change effect, not an arbitrary spread added to the forward rate.*

Concrete Example: A Few Basis Points Can Matter

- ▶ Consider a delayed floating coupon with

$$N = 100 \text{ million}, \quad \alpha = 0.50,$$

paid at T_{pay} , with

$$P(t, T_{\text{pay}}) = 0.97.$$

- ▶ Suppose the ordinary forward rate is

$$L(t; T_{\text{st}}, T_{\text{end}}) = 4.00\%,$$

while the model-implied delayed-payment expectation is

$$\mathbb{E}^{\mathbb{Q}^{T_{\text{pay}}}} [L(T; T_{\text{st}}, T_{\text{end}})] = 4.03\%.$$

- ▶ The convexity adjustment is therefore

$$\text{CA} = 3 \text{ bp.}$$

- ▶ The valuation impact is

$$\begin{aligned} \Delta V &= N \alpha P(t, T_{\text{pay}}) \text{CA} \\ &\approx 100\text{m} \times 0.50 \times 0.97 \times 0.0003 \\ &\approx \boxed{14,550}. \end{aligned}$$

Lesson

- ▶ *A small rate adjustment can become material on large notionals or across many cash flows.*

When the Effect Becomes Important

- ▶ The convexity effect tends to matter more when there is:
 - ▶ a longer delay between T_{end} and T_{pay} ;
 - ▶ higher interest-rate volatility;
 - ▶ stronger dependence between the fixing rate and relevant discount factors;
 - ▶ large notionals or repeated delayed coupons;
 - ▶ nonlinear products whose payoff amplifies rate movements.
- ▶ A robust implementation should:
 - ▶ identify the fixing, accrual-end and payment dates separately;
 - ▶ choose the correct pricing numeraire;
 - ▶ derive or model the required measure-change adjustment;
 - ▶ compare the adjusted value with the naive forward-rate approximation;
 - ▶ stress volatility and correlation assumptions.

Important

- ▶ *If $T_{\text{pay}} = T_{\text{end}}$, the standard forward-measure simplification applies. The convexity issue arises from the mismatch between the rate's natural maturity and the actual payment date.*

Learn more

- ▶ For market measures, payment delays and convexity adjustments, see [Lecture FE11: Market Model and Convexity Adjustments](#).

Quant Rule #27

Whenever a floating rate is paid away from its natural maturity date:

- ▶ keep fixing, accrual-end and payment dates distinct;
- ▶ identify the measure under which the rate is a martingale;
- ▶ identify the numeraire associated with the actual payment date;
- ▶ quantify the resulting convexity adjustment;
- ▶ validate its sensitivity to volatility and correlation.

Quant Rule #27

- ▶ A delayed payment changes the numeraire used to value the cash flow. Do not assume that the ordinary forward rate remains the correct expectation under the payment-date measure.

same fixing + different payment date $\neq$ same valuation formula.

Final message

- ▶ *Payment timing is part of the stochastic valuation problem, not merely a calendar convention.*

Quant Mistake #28: Pricing FX Forwards without Cross-Currency Basis Spreads

The incorrect belief

- ▶ Once we know the spot FX rate and the domestic and foreign yield curves, the FX forward is completely determined.

The mistake

- ▶ *In the market, the classical relation between spot FX, interest rates and FX forwards does not always hold exactly.
The cost of obtaining one currency through the FX market can differ from what the two standalone yield curves imply.
This difference is reflected in the **cross-currency basis spread**.*

Ignoring the basis can lead to:

- ▶ incorrect FX forward points;
- ▶ incorrect cross-currency swap values;
- ▶ incorrect hedges against currency funding risk.

spot FX + two yield curves $\neq$ always enough.

The Classical FX Forward Relation

- ▶ Let

$$y(t) := y_f^d(t)$$

denote the FX rate in units of domestic currency per unit of foreign currency.

- ▶ Let

$$P_d(t, T), \quad P_f(t, T)$$

denote the domestic and foreign zero-coupon bonds.

- ▶ In the classical no-basis setting,

$$y_F(t, T) = y(t) \frac{P_f(t, T)}{P_d(t, T)}.$$

- ▶ For constant interest rates,

$$P_d(t, T) = e^{-r_d(T-t)}, \quad P_f(t, T) = e^{-r_f(T-t)},$$

so that

$$y_F(t, T) = y(t) e^{(r_d - r_f)(T-t)}.$$

Classical interpretation

- ▶ *The FX forward is determined by spot FX and the difference between domestic and foreign interest rates.*

Why Does the Cross-Currency Basis Appear?

- ▶ The classical relation assumes that funding the two currencies through the FX market is consistent with their standalone interest-rate curves.
- ▶ In practice, demand for funding can be very different across currencies.
- ▶ For example, if many institutions want to obtain **USD funding** through the FX market, the cost of obtaining dollars through EUR/USD swaps can increase.
- ▶ This does not necessarily move the whole domestic USD yield curve.
- ▶ Instead, the difference appears in FX swaps and cross-currency swaps as the

cross-currency basis.

Key idea

- ▶ *The basis reflects the difference between the cost of obtaining a currency through the FX market and the cost implied by the standalone yield curves.*

The Market Forward Can Differ from the Classical Forward

- ▶ Without basis, we would expect

$$y_F(t, T) = y(t) \frac{P_f(t, T)}{P_d(t, T)}.$$

- ▶ In practice, the observed FX forward may satisfy

$$y_F(t, T) \neq y(t) \frac{P_f(t, T)}{P_d(t, T)}.$$

- ▶ We introduce basis-spread factors

$$ZF_d(t, T), \quad ZF_f(t, T).$$

- ▶ The basis-adjusted relation becomes

$$y_F(t, T) = y(t) \frac{P_f(t, T) ZF_f(t, T)}{P_d(t, T) ZF_d(t, T)}.$$

- ▶ The factors ZF_d and ZF_f are bootstrapped from FX and cross-currency market instruments.

Key point

- ▶ *The cross-currency basis contains additional market information that is not present in the two standalone yield curves.*

Concrete Example: Ignoring Basis Moves the FX Forward

- Suppose

$$y(0) = 1.10.$$

- Let the one-year domestic and foreign discount factors be

$$P_d(0, 1) = e^{-0.05}, \quad P_f(0, 1) = e^{-0.03}.$$

- Ignoring the basis gives

$$y_F^{\text{no basis}}(0, 1) = 1.10 \frac{e^{-0.03}}{e^{-0.05}} \\ \approx 1.1222.$$

- Suppose the cross-currency market implies

$$ZF_d(0, 1) = 1, \quad ZF_f(0, 1) = e^{-0.005}.$$

- Including the basis gives

$$y_F^{\text{basis}}(0, 1) = 1.10 \frac{e^{-0.03} e^{-0.005}}{e^{-0.05}} \\ \approx 1.1166.$$

- The difference is approximately

56 pips.

Lesson

- A relatively small cross-currency basis can produce a visible change in FX forward points.*

How Is the Basis Used in Practice?

- ▶ FX and cross-currency swap markets provide **basis quotes** for different maturities.
- ▶ The market therefore provides a term structure of basis spreads, rather than one constant spread for all maturities.
- ▶ These quotes are used to bootstrap the corresponding basis factors

$$ZF_d(t, T), \quad ZF_f(t, T).$$

- ▶ The FX forward is then obtained from

$$y_F(t, T) = y(t) \frac{P_f(t, T)ZF_f(t, T)}{P_d(t, T)ZF_d(t, T)}.$$

Important

- ▶ *The cross-currency basis is not simply added to the FX forward. It is used to adjust the relationship between the domestic and foreign yield curves.*

Quant Rule #28

Before pricing an FX forward:

- ▶ identify the domestic and foreign currencies;
- ▶ obtain the spot FX rate;
- ▶ construct the domestic and foreign yield curves;
- ▶ check the relevant FX and cross-currency basis quotes;
- ▶ include the basis when constructing the FX forward curve.

Quant Rule #28

- ▶ Do not assume that spot FX and the two standalone yield curves always fully determine the market FX forward.
Check the [cross-currency basis](#).

FX market = spot + interest-rate differential + cross-currency effect.

Final message

- ▶ *When the market trades a cross-currency basis, that basis contains information that is missing from the two standalone yield curves.*

Learn more

- ▶ For FX forwards and cross-currency basis spreads, see [Lecture FE10: Foreign Exchange \(FX\) and Inflation](#).

Quant Mistake #29: Taking Positive Exposure Before Netting-Set Aggregation

The incorrect belief

- Compute positive exposure for each trade separately and then add the results:

$$E(t) = \sum_{j=1}^m \max(V_j(t), 0).$$

The mistake

- *If trades belong to the same legally enforceable netting set, negative market values can offset positive market values upon counterparty default. Trade values must therefore be aggregated within the netting set before taking the positive part.*

$$\max\left(\sum_j V_j, 0\right) \neq \sum_j \max(V_j, 0)$$

in general.

Exposure Is Defined at the Netting-Set Level

- ▶ Let

$$V_j(t)$$

be the future mark-to-market of trade j .

- ▶ For a legally enforceable netting set n , define

$$V_n(t) = \sum_{j \in n} V_j(t).$$

- ▶ Ignoring collateral for the moment, counterparty exposure for that netting set is

$$E_n(t) = \max(V_n(t), 0) = \max\left(\sum_{j \in n} V_j(t), 0\right).$$

Key point

- ▶ *Netting determines the amount owed after legally permitted offsets between trades.
The positive-part operator is then applied to that net amount.*

Concrete Example: Taking Positive Exposure Too Early

- Suppose that at one future scenario date two trades in the same netting set have values

$$V_1 = +10 \text{ million}, \quad V_2 = -8 \text{ million}.$$

- The correct netted exposure is

$$\begin{aligned} E_{\text{correct}} &= \max(10 - 8, 0) \\ &= \boxed{2 \text{ million}}. \end{aligned}$$

- If the positive part is applied trade by trade first,

$$\begin{aligned} E_{\text{wrong}} &= \max(10, 0) + \max(-8, 0) \\ &= \boxed{10 \text{ million}}. \end{aligned}$$

Result

- Taking positive exposure too early destroys the offset from the negative trade. Here the exposure is overstated by a factor of five.*

Why the Error Is Systematically One-Sided

- ▶ For any set of trade values,

$$\left(\sum_j v_j \right)^+ \leq \sum_j v_j^+.$$

- ▶ Thus taking the positive part trade by trade can only preserve or increase exposure relative to proper netting-set aggregation.
- ▶ Taking expectations gives

$$\mathbb{E} \left[\left(\sum_j v_j(t) \right)^+ \right] \leq \sum_j \mathbb{E} [v_j(t)^+].$$

- ▶ The size of the netting benefit depends on:
 - ▶ correlation between trade values;
 - ▶ portfolio composition and maturities;
 - ▶ which trades legally belong to the same netting set.

Central lesson

- ▶ *Taking positive exposure too early removes legally permitted offsets between positive and negative trade values.*

The Error Propagates Directly into CVA

- ▶ In a simplified setting without exposure–default dependence, unilateral CVA may be written schematically as

$$\text{CVA} \approx (1 - R) \sum_k D(0, t_k) \text{EE}(t_k) \Delta PD_k,$$

where

$$\text{EE}(t_k) = \mathbb{E}^{\mathbb{Q}} [E(t_k)].$$

- ▶ If one computes instead

$$E_{\text{wrong}}(t) = \sum_j V_j(t)^+,$$

expected exposure is overstated and, all else equal, so is CVA.

- ▶ The same aggregation error can affect:
 - ▶ PFE and exposure profiles;
 - ▶ economic capital and counterparty limits;
 - ▶ collateral and funding calculations;
 - ▶ wrong-way-risk analysis.

Practical warning

- ▶ *A sophisticated Monte Carlo engine cannot repair an exposure definition built at the wrong aggregation level.*

Netting Must Follow the Legal Agreement

- ▶ The correct rule **is not** “net everything with the same counterparty.”
- ▶ Trades may belong to different:
 - ▶ master agreements or legally enforceable netting sets;
 - ▶ collateral agreements;
 - ▶ legal entities;
 - ▶ cleared and bilateral portfolios.
- ▶ For each netting set n , first compute

$$V_n(t) = \sum_{j \in n} V_j(t).$$

- ▶ Then apply the collateral terms associated with that netting set. Schematically,

$$E_n(t) = \max(V_n(t) - C_n(t), 0),$$

where $C_n(t)$ denotes the relevant collateral balance.

Correct principle

- ▶ *Aggregation follows legal enforceability first.*
Collateral treatment is applied at the appropriate netting-set level, and positive exposure is determined only afterwards.

Learn more

- ▶ For counterparty exposure, netting and valuation adjustments, see [Lecture FE12: Valuation Adjustments- xVA \(CVA, BCVA and FVA\)](#).

Quant Rule #29

When computing counterparty exposure:

- ▶ identify the legally enforceable netting sets;
- ▶ aggregate simulated trade values within each netting set;
- ▶ apply the relevant collateral and margin rules;
- ▶ only then take the positive exposure;
- ▶ compute EE, PFE and CVA from the resulting net exposure.

Quant Rule #29

- ▶ Net first. Take positive exposure second.

Applying the positive part trade by trade destroys offsetting negative values and systematically overstates exposure relative to legally permitted netting.

$$\left(\sum_{j \in n} v_j \right)^+ \neq \sum_{j \in n} v_j^+ \quad \text{in general.}$$

Final message

- ▶ *Exposure is a property of the legally enforceable netting set, not of each trade in isolation.*

Quant Mistake #30: Treating Value-at-Risk as the Maximum Possible Loss

The incorrect belief

- ▶ A one-day 99% VaR of \$10 million means the portfolio cannot lose more than \$10 million in one day.

The mistake

- ▶ *Value-at-Risk is a quantile of the loss distribution- not an upper bound on losses.*

If L denotes portfolio loss, then

$$\text{VaR}_\alpha(L) = \inf \{ \ell : \Pr(L \leq \ell) \geq \alpha \}.$$

At confidence level $\alpha = 99\%$, VaR identifies a threshold exceeded only in the worst roughly 1% of cases.

VaR tells us where the tail starts, not how far the tail can go.

What a 99% VaR Actually Means

- ▶ Suppose

$$\text{VaR}_{99\%} = 10 \text{ million.}$$

- ▶ For a continuous loss distribution, this means approximately

$$\Pr(L > 10 \text{ million}) = 1\%.$$

- ▶ It does not imply

$$L \leq 10 \text{ million.}$$

- ▶ Losses in that worst 1% could be

11m, 30m, 100m,

or even larger, depending on the portfolio and assumed loss distribution.

Key distinction

- ▶ *VaR describes the location of a percentile.
It says almost nothing by itself about the severity of losses beyond that percentile.*

Example: Same VaR, Completely Different Tail Risk

- ▶ Consider two simplified one-day loss distributions:

	99% of days	1% of days
Portfolio A	0 to 10m	11m
Portfolio B	0 to 10m	100m

- ▶ Both can have approximately

$$\text{VaR}_{99\%} = 10 \text{ million.}$$

- ▶ But their losses beyond VaR are radically different:

11 million versus 100 million.

Lesson

- ▶ *VaR can assign the same headline risk number to portfolios with very different extreme-loss profiles.*

Expected Shortfall Looks Beyond the VaR Threshold

Value-at-Risk

- Where does the extreme-loss tail begin?

$$\text{VaR}_\alpha .$$

Expected Shortfall

- How large are losses on average once we are in the extreme tail?

$$\text{ES}_\alpha = \mathbb{E}[L \mid L > \text{VaR}_\alpha]$$

for a continuous loss distribution.

For the previous stylized example:

	$\text{VaR}_{99\%}$	$\text{ES}_{99\%}$
Portfolio A	10m	$\approx 11\text{m}$
Portfolio B	10m	$\approx 100\text{m}$

Important

- *Expected Shortfall adds information about tail severity, but it is still model- and scenario-dependent- not a maximum-loss guarantee.*

VaR Can Look Better while Tail Risk Gets Worse

- ▶ Suppose a position earns a small premium in almost every scenario but suffers a very large loss in rare states.
- ▶ If the rare loss lies beyond the chosen VaR confidence level, it may have little or no effect on the reported VaR.
- ▶ For example:

probability	one-day P&L
99%	+0.1m
1%	−50m

- ▶ At the 99% quantile, VaR can look very small even though the portfolio carries a severe crash exposure just beyond the cutoff.

Risk-management warning

- ▶ *Optimizing a portfolio only to reduce VaR can encourage risk to migrate just beyond the chosen quantile.*

VaR Must Be Used with Other Risk Measures

A robust risk framework should not ask only

“What is the 99% VaR?”

It should also examine:

- ▶ Expected Shortfall and other tail measures;
- ▶ historical and hypothetical stress scenarios;
- ▶ concentration, liquidity and jump risk;
- ▶ sensitivity to the horizon and confidence level;
- ▶ sensitivity to distributional assumptions and historical window;
- ▶ backtesting against realized P&L.

Practical principle

- ▶ *VaR is one summary of a loss distribution.
It is not a substitute for understanding the entire tail and the scenarios that can generate catastrophic losses.*

Learn more

- ▶ For market-risk measurement, VaR, Expected Shortfall and stress testing, see [Lecture FE13: Value-at-Risk and Expected Shortfall](#).

Quant Rule #30

Whenever VaR is reported, ask:

- ▶ What confidence level and time horizon are being used?
- ▶ What loss model, data window and assumptions generate the distribution?
- ▶ What happens beyond the VaR threshold?
- ▶ What is the corresponding Expected Shortfall?
- ▶ Which stress scenarios are not represented by the model or historical sample?

Quant Rule #30

- ▶ Value-at-Risk is a percentile of the loss distribution, not the maximum possible loss.

Always investigate the losses beyond VaR instead of treating the VaR number as a worst-case bound.

VaR measures a threshold; risk management must understand the tail.

Final message

- ▶ *The most dangerous losses are precisely the ones that VaR does not describe.*